原和恢 SAP HANA 数据

Snap Creator Framework

NetApp
October 27, 2021

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□原和恢□ SAP HANA 数据□

□可以使用 SAP HANA Studio 和 Snap Creator □原和恢□ SAP HANA 数据□。

1. 在 SAP HANA Studio 中：
   a. □ SAP HANA 系□□□恢□。
   b. SAP HANA 系□已□□。
   c. □□恢□□型。
   d. 提供日志□□位置。
   e. 此□将□示数据□□列表
   f. □□ backup 可□看外部□□ ID 。
2. □于□□行集群模式 Data ONTAP 的存□系□：
   a. 只有在使用最新□□以外的任何其他□□□行□原□才需要此□□。
   b. □主存□中的 "`Volume SnapRestore` " 才需要此功能。
   c. 停用 SnapVault □系
3. 在 Snap Creator 中：
   a. □ SAP HANA 系□□□ "`Restore` " 。
   b. 根据主存□□□的可用性，□□从主存□或二□存□□原。
   c. □□存□控制器，卷名称和 Snapshot 副本名称。Snapshot 副本名称与先前的□□ ID 相□。
   d. □于多□点 SAP HANA 系□，需要□原多个卷：
      i. □□ * 添加更多□原□ * 。
      ii. □□存□控制器，卷名称和 Snapshot 副本名称。
      iii. □所有所需卷重□此□程。
   e. □于多租□数据□容器（ MDC ）□租□数据□系□，系□和租□数据□都将□原。
   f. □原□程已□□
   g. 已完成所有卷的□原。
4. 在数据□□点上，卸□并挂□所有数据卷以清理 " □旧的 NFS 句柄 " 。
5. 在 SAP HANA Studio 中：
   a. 在□□列表中□□ * 刷新 * 。
   b. □□可用于恢□的□□（□色□目）。
   c. □□恢□□程。
   d. □于多租□数据□容器（ MDC ）□租□数据□系□，首先□系□数据□□□恢□□程，然后□租□数据□□□恢□□程。
   e. SAP HANA 系□已□□。
6. （可□）恢□所有已□原卷的 SnapVault □系。

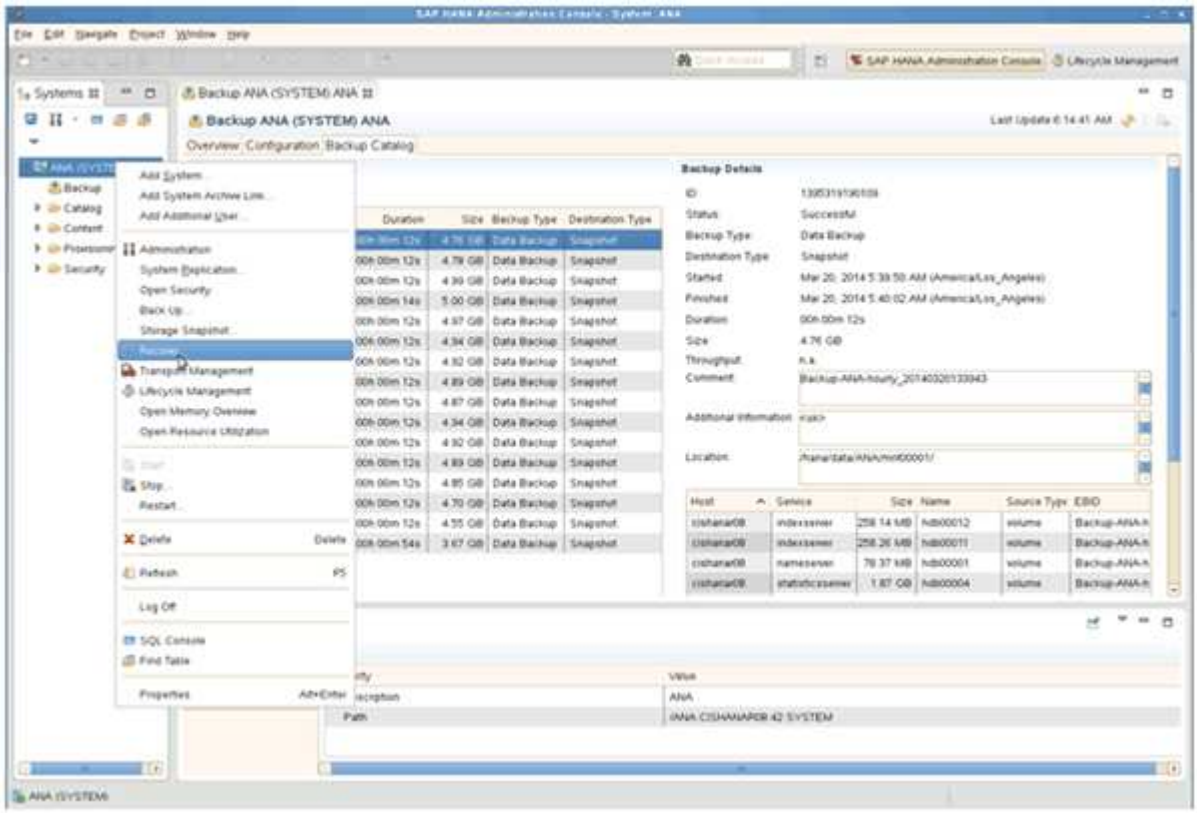
在存□系□中，只有在使用最新□□以外的□□□行□原□，才需要□行此□□。

从主存□□原和恢□数据□

□可以从主存□□原和恢□数据□。

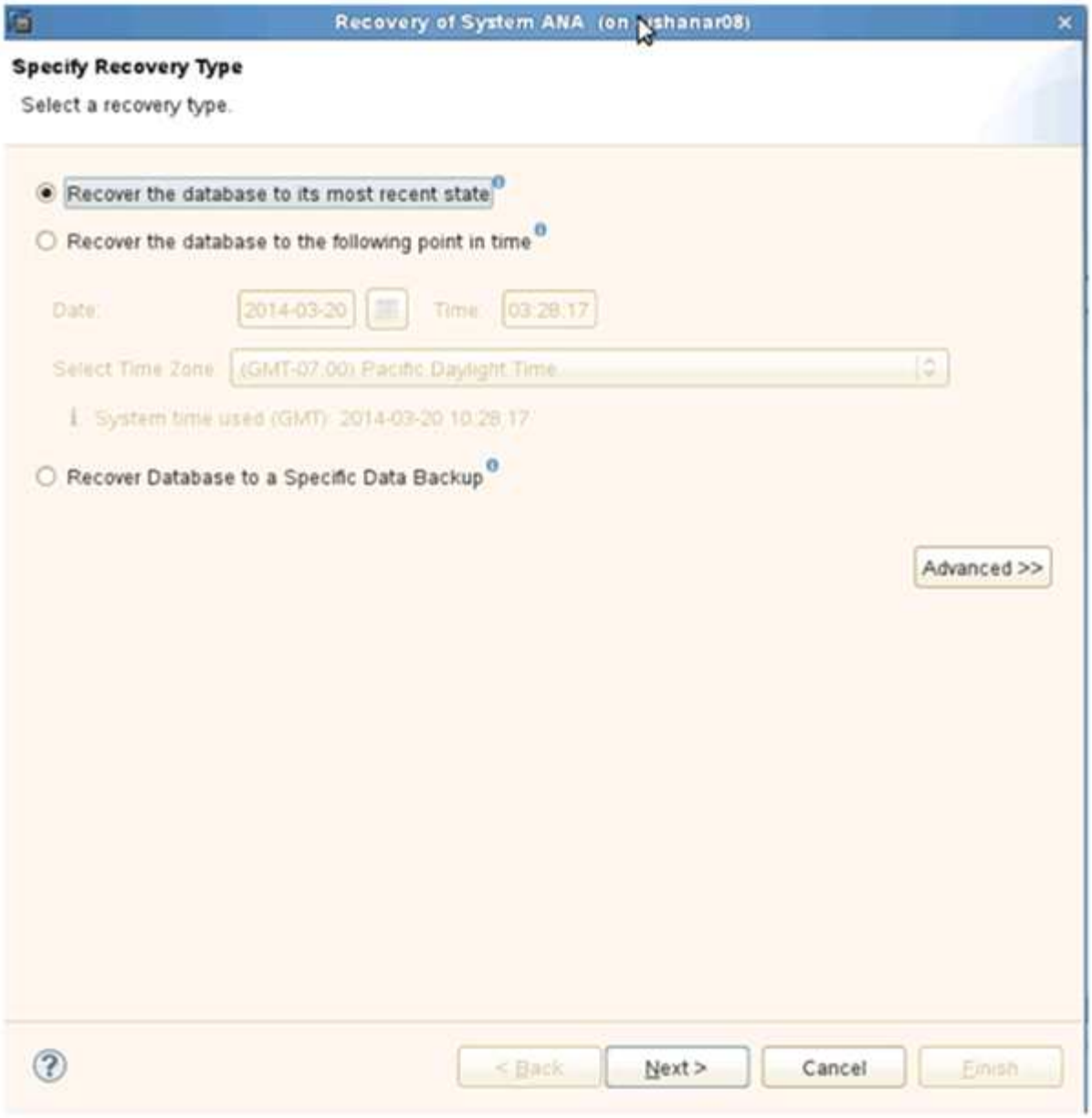
□不能从 Snap Creator □原基于文件的□□副本。

1. 在 SAP HANA Studio 中，□ SAP HANA 系□□□ * 恢□ * 。

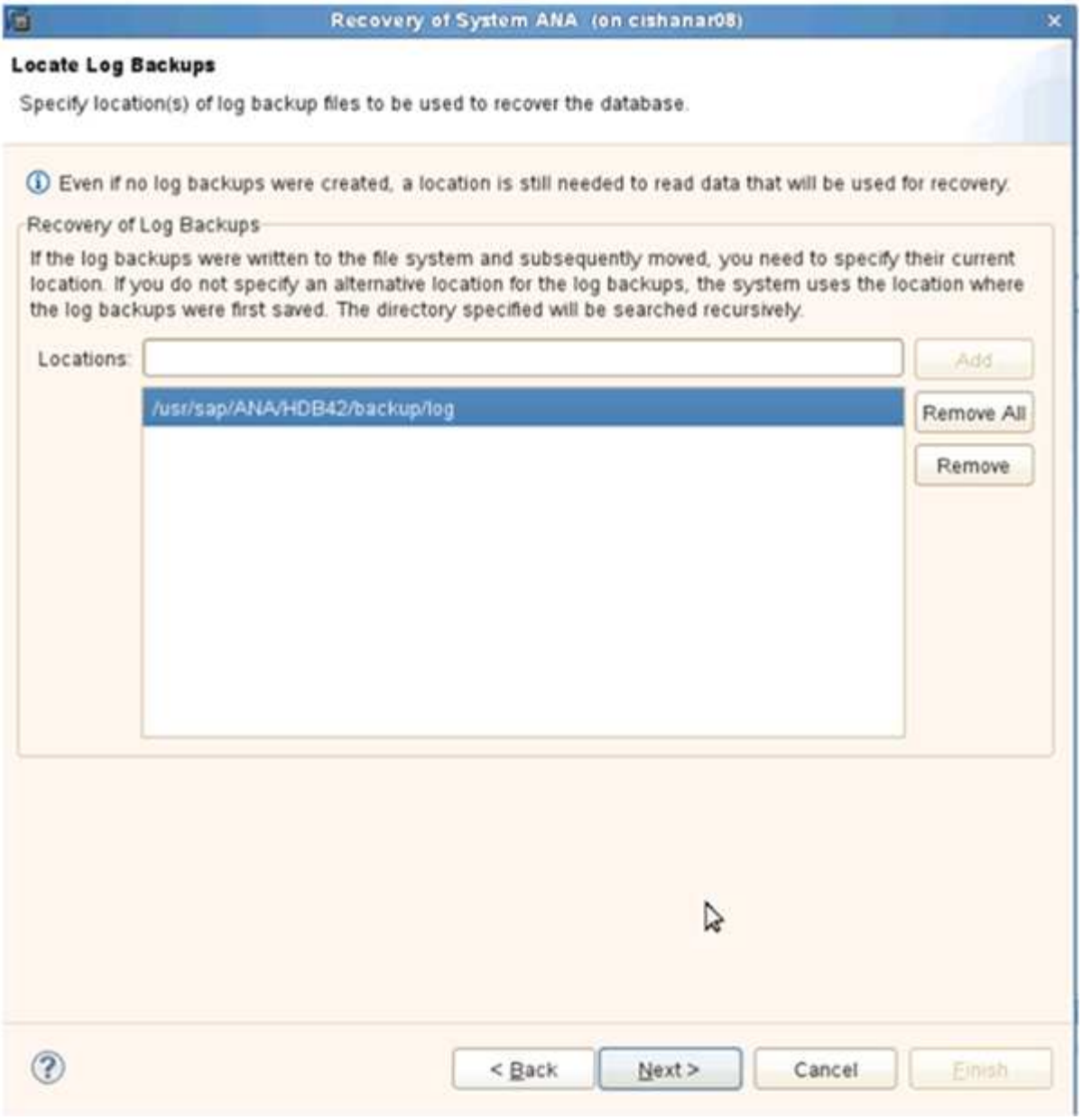


SAP HANA 系□□□。

2. □□恢□□型并□□ * 下一□ * 。

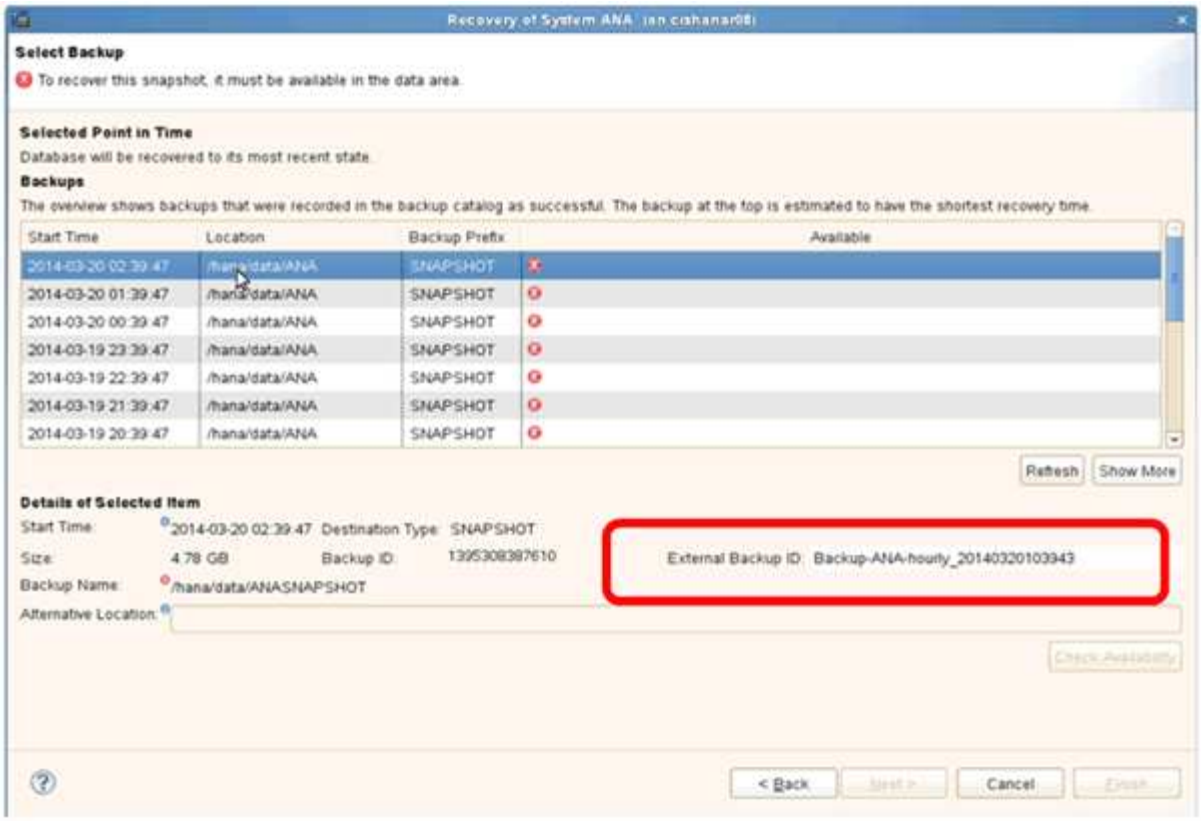


3. 提供日志□□位置，然后□□ * 下一□ * 。



□看到的可用□□列表取决于□□目□的内容。

4. □□所需的□□并□□外部□□ ID 。



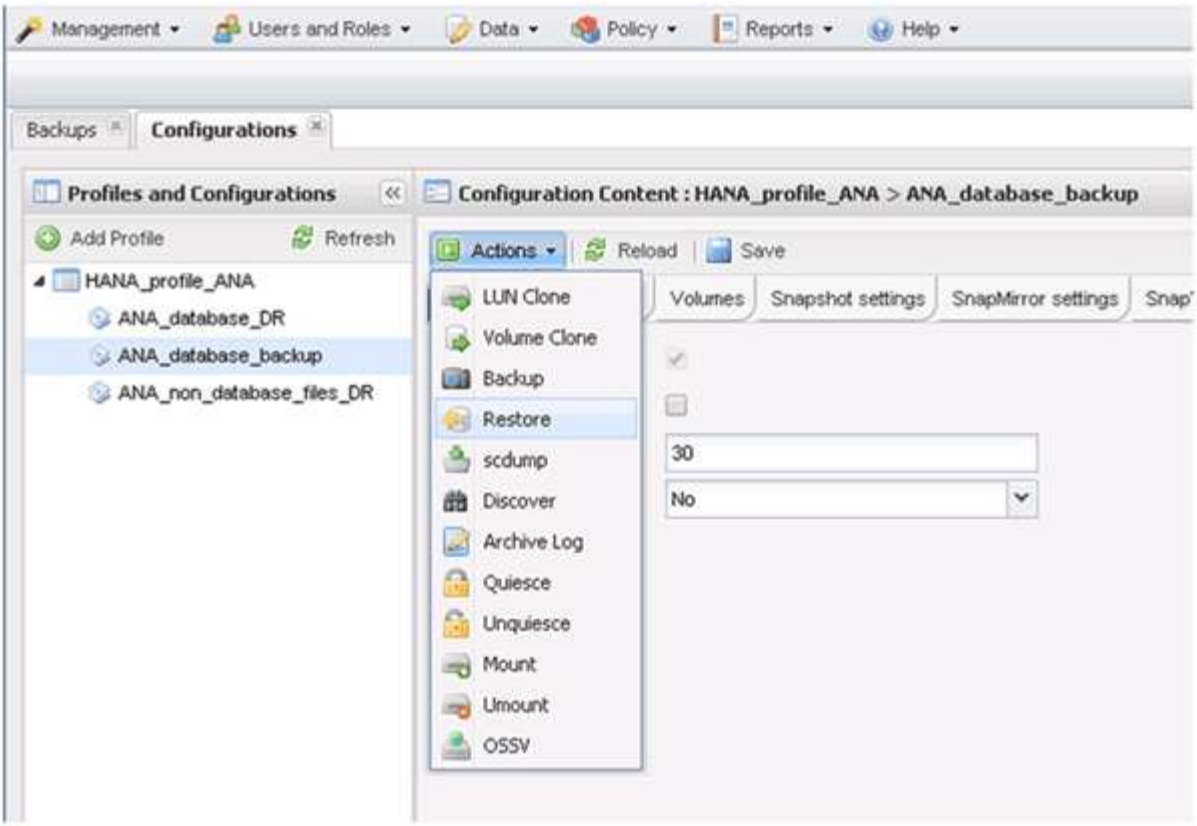
5. 停用 SnapVault □系。

> ⓘ 只有集群模式 Data ONTAP 才需要□行此□□。

如果需要□原的 Snapshot 副本早于当前用作 SnapVault 基本 Snapshot 副本的 Snapshot 副本，□必□先在集群模式 Data ONTAP 中停用 SnapVault □系。□此，□在□□集群控制台上□行以下命令：

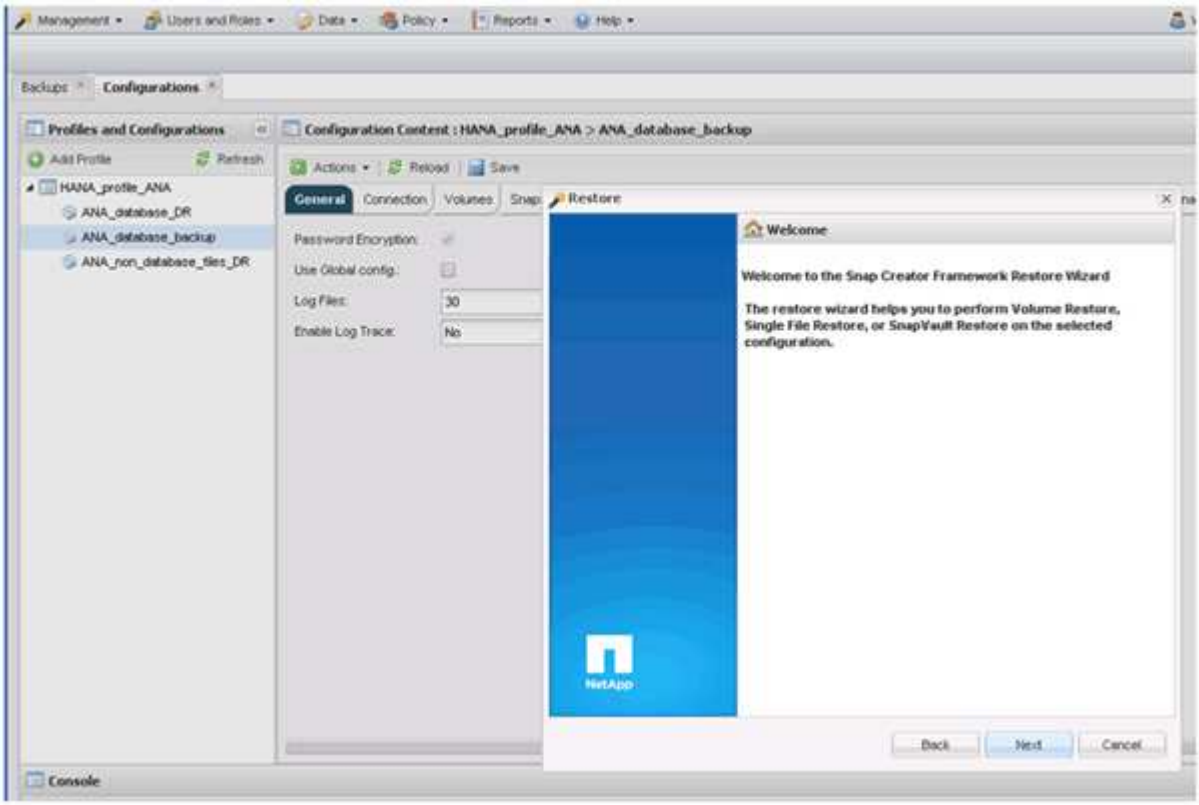
```
hana::> snapmirror quiesce -destination-path hana2b:backup_hana_data
Operation succeeded: snapmirror quiesce for destination
hana2b:backup_hana_data.

hana::> snapmirror delete -destination-path hana2b:backup_hana_data
Operation succeeded: snapmirror delete the relationship with destination
hana2b:backup_hana_data.

hana::> snapmirror release -destination-path hana2b:backup_hana_data
[Job 6551] Job succeeded: SnapMirror Release Succeeded
```

6. 在 Snap Creator GUI 中，□□ SAP HANA 系□，然后□□ * 操作 * > * □原 * 。

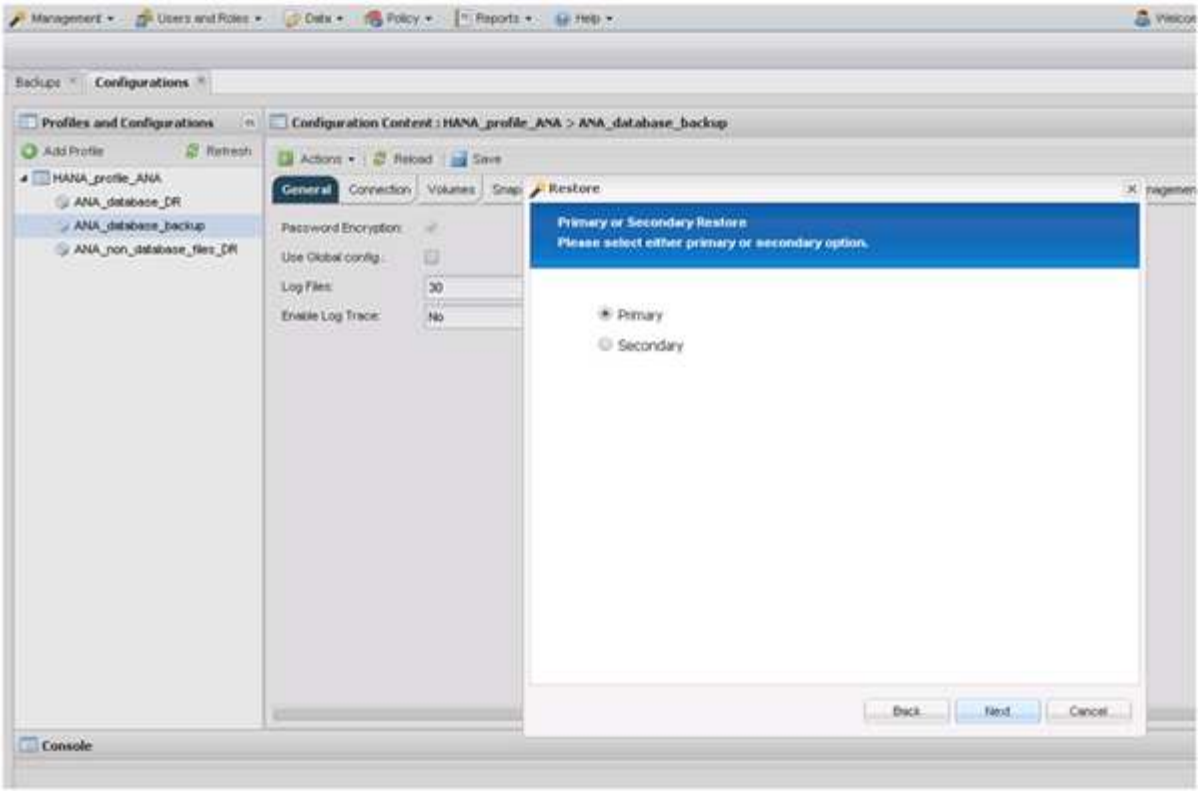


此□将□示 Welcome to the Snap Creator Framework Restore Wizard 屏幕。

7. □□ * 下一□ * 。

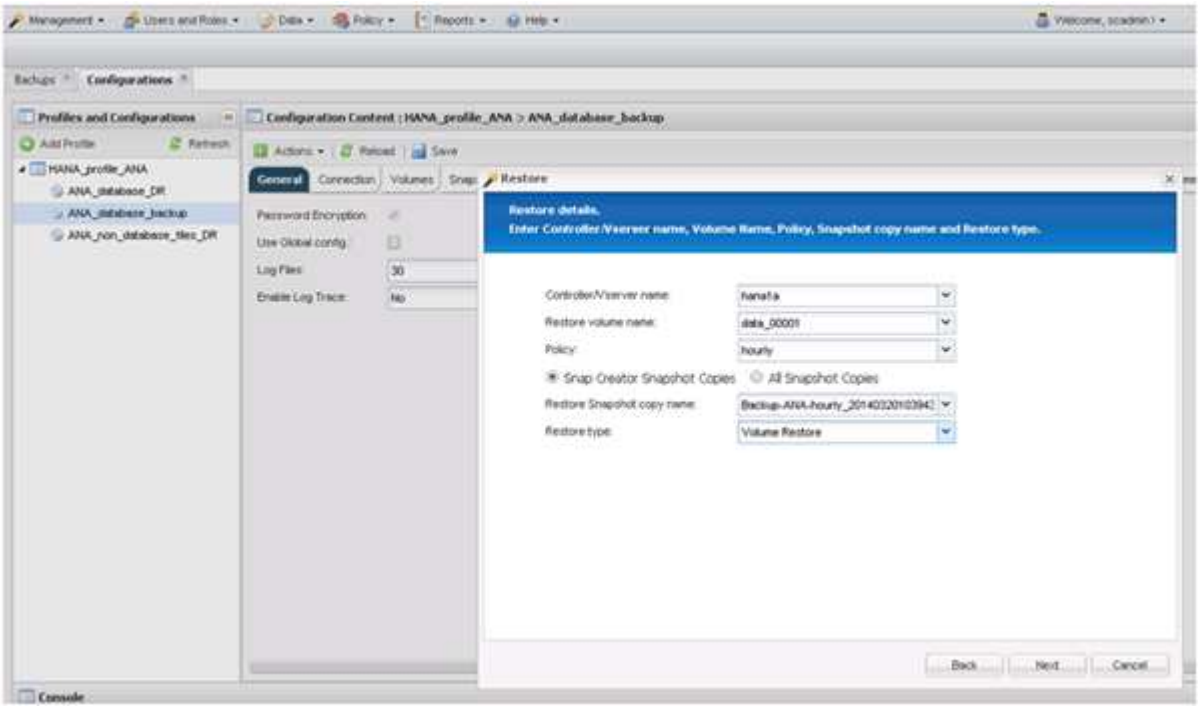


8. □□ * 主 * 并□□ * 下一□ * 。

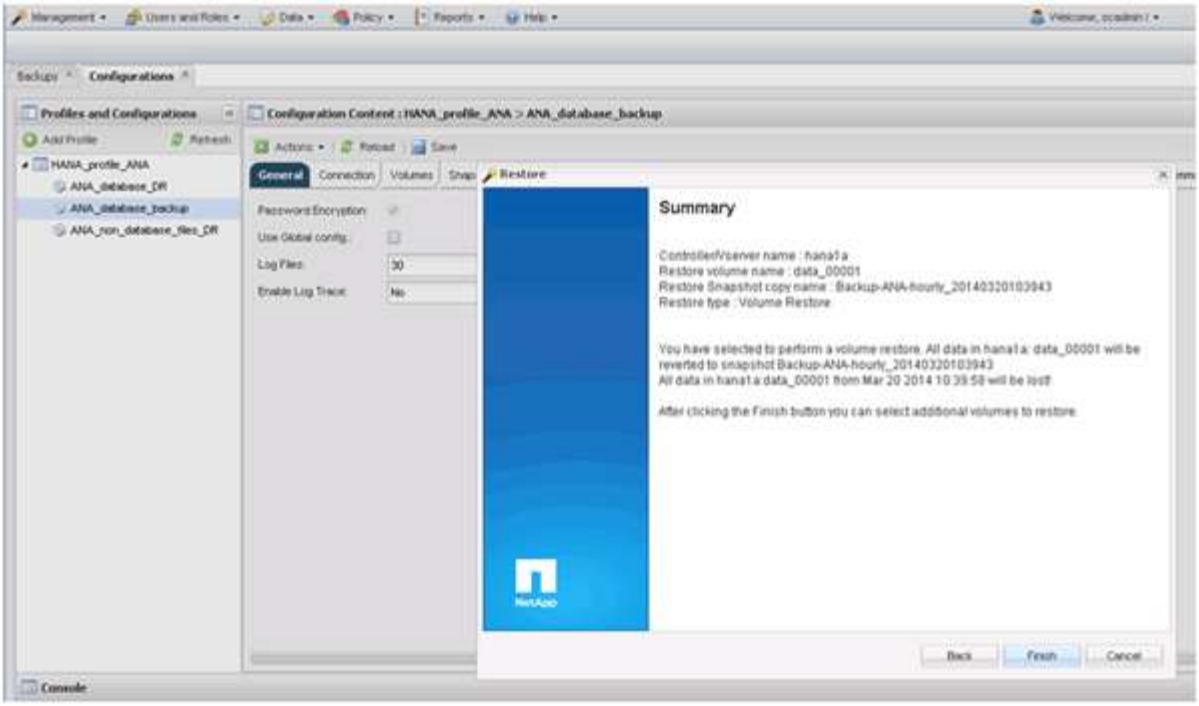


9. □□从主存□□原。
10. □□存□控制器，卷名称和 Snapshot 名称。

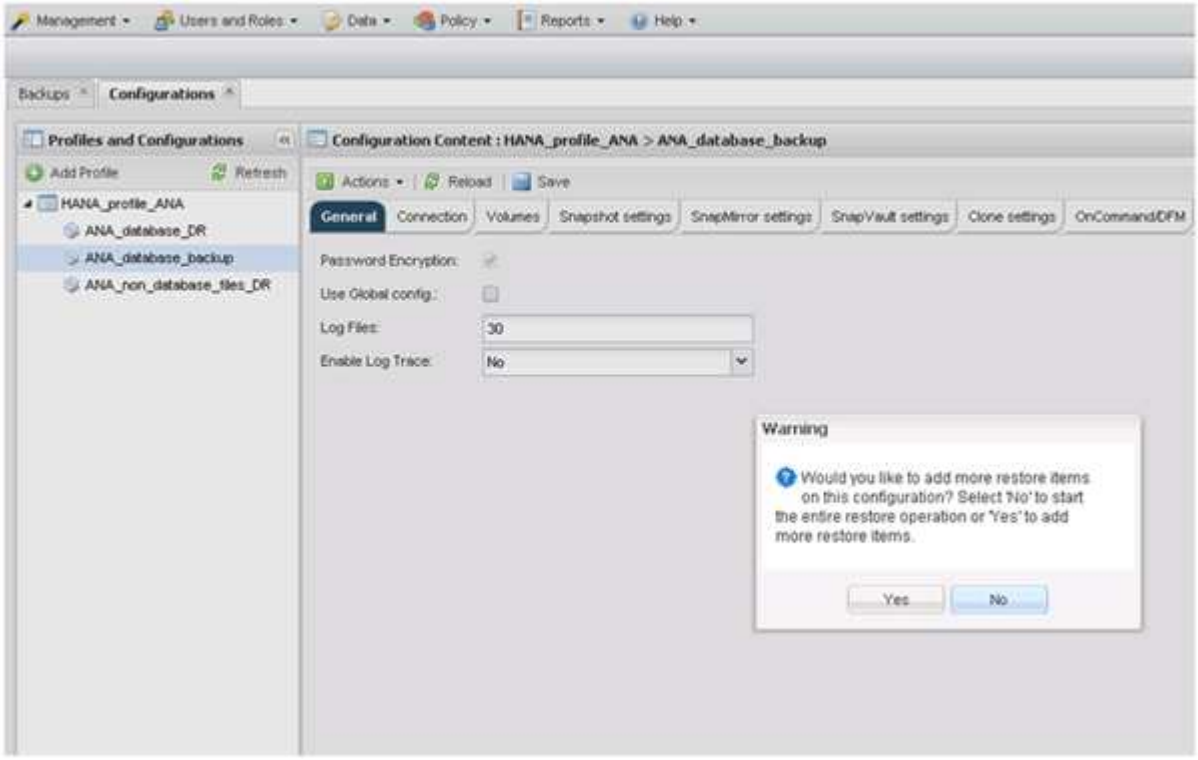
    Snapshot 名称与在 SAP HANA Studio 中□□的□□ ID □□。



11. □□ * 完成 * 。

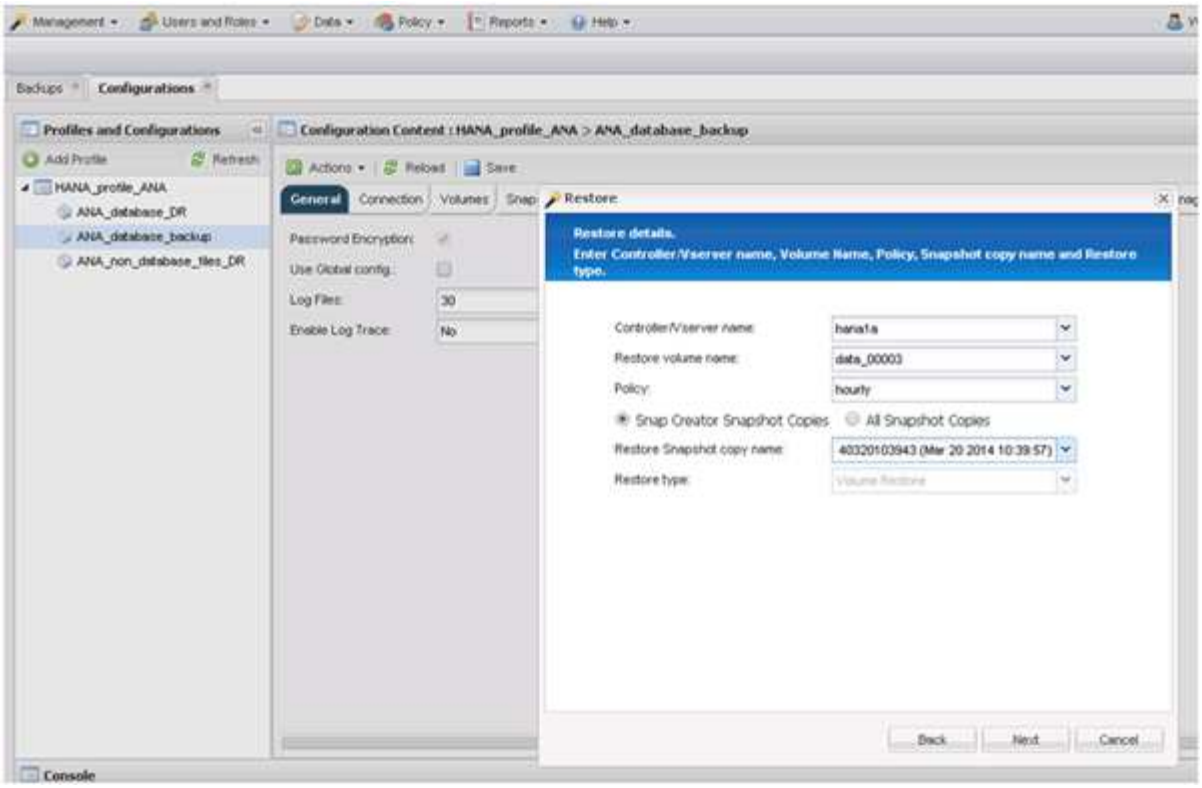


12. □□ * 是 * 添加更多□原□。

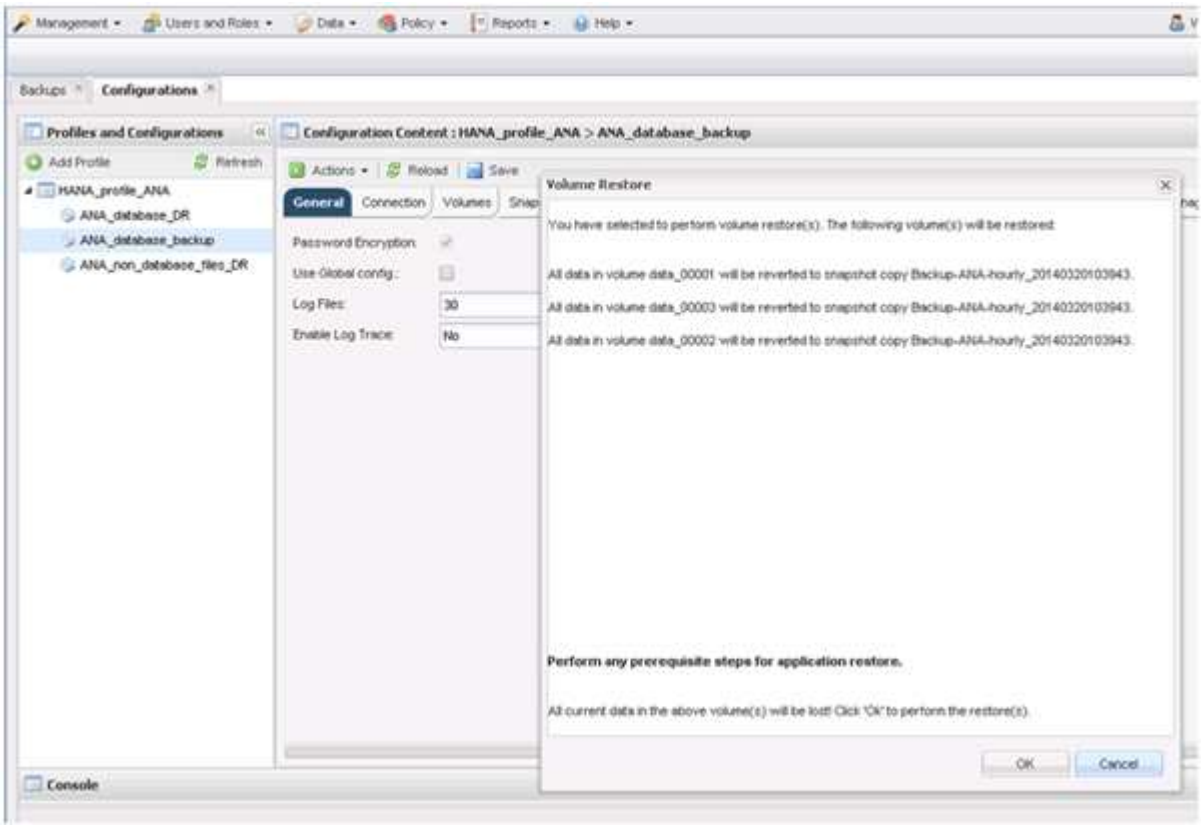


13. □□存□控制器，其他卷名称和 Snapshot 名称。

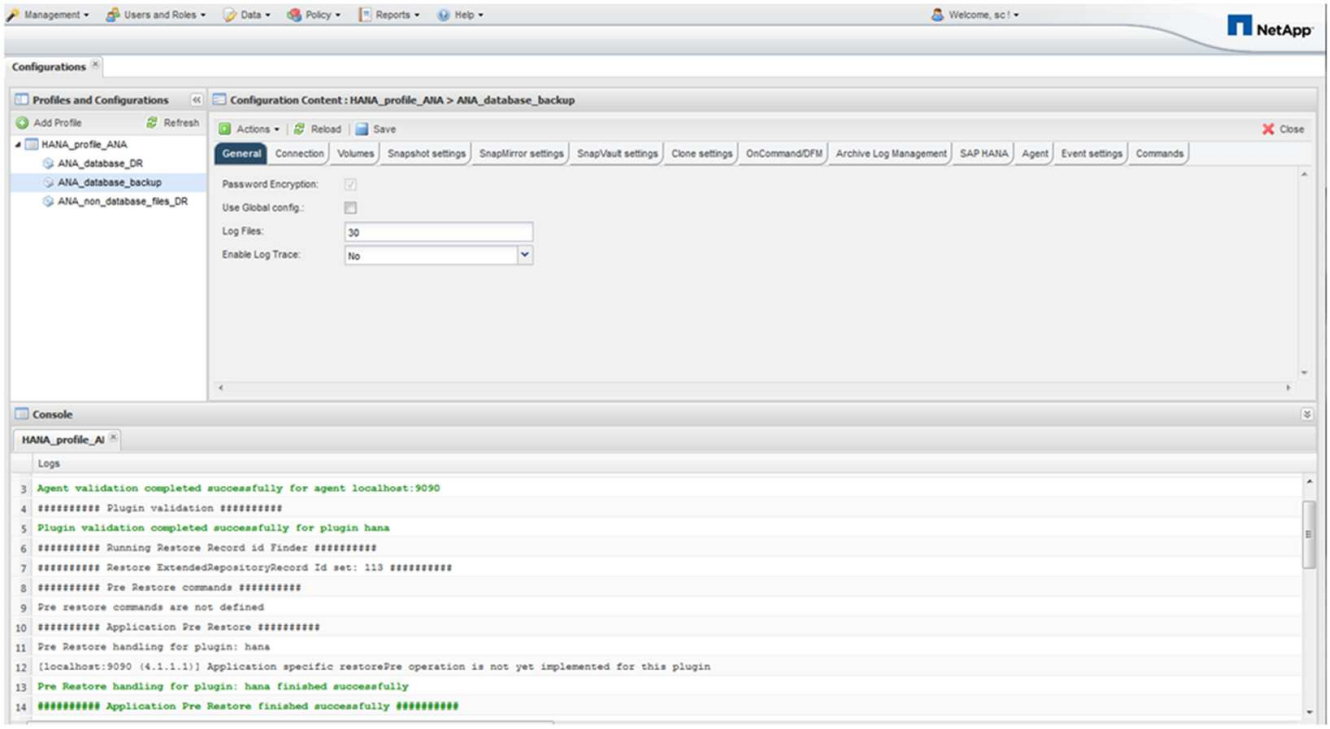
    Snapshot 名称与在 SAP HANA Studio 中□□的□□ ID □□。



14. 重□□□ 10 到 13， 直到添加了所有必需的卷；在本示例中，需要□□原□程□□ data_00001， data_00002 和 data_00003 。
15. □□所有卷后，□□ * □定 * 以□□□原□程。



此□将□□□原□程。



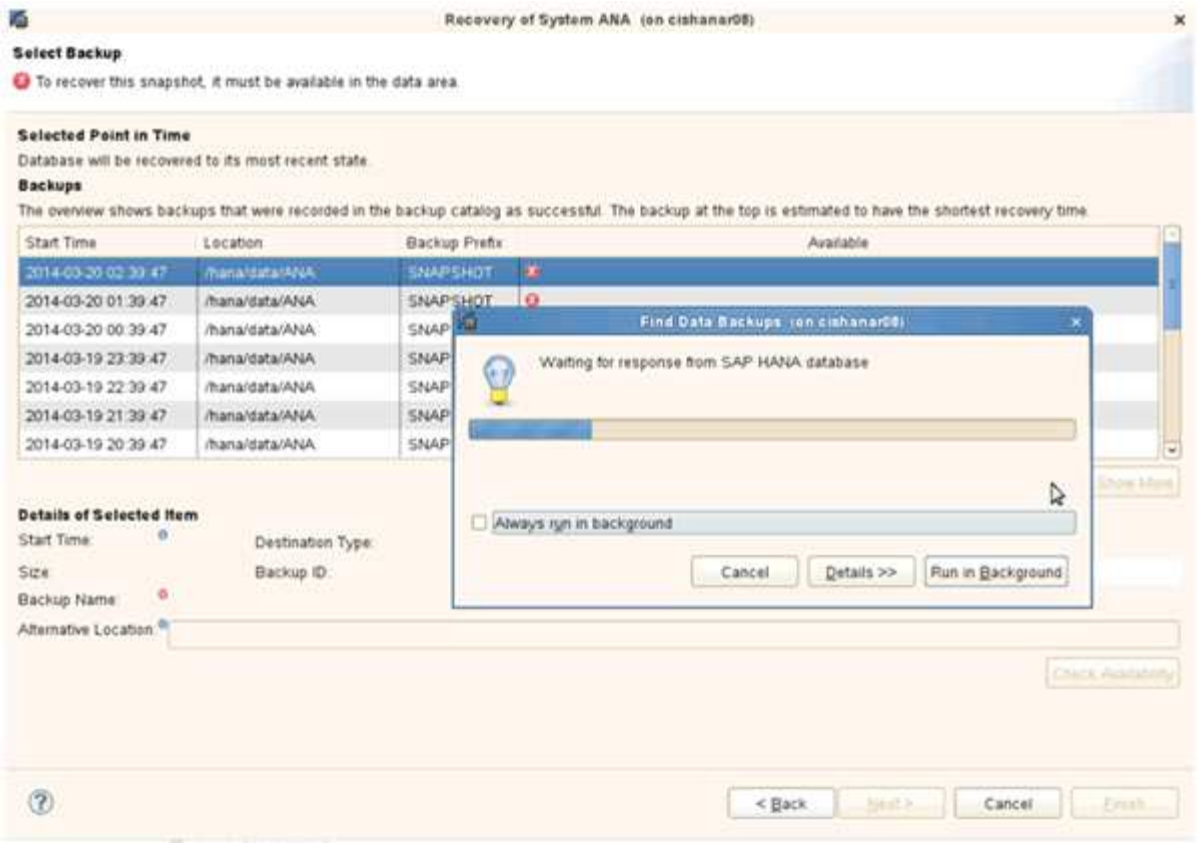
□等待□原□程完成。

16. 在□个数据□□点上，重新挂□所有数据卷以清除□旧的 NFS 句柄。

    在此示例中，所有三个卷都需要在□个数据□□点上重新挂□。

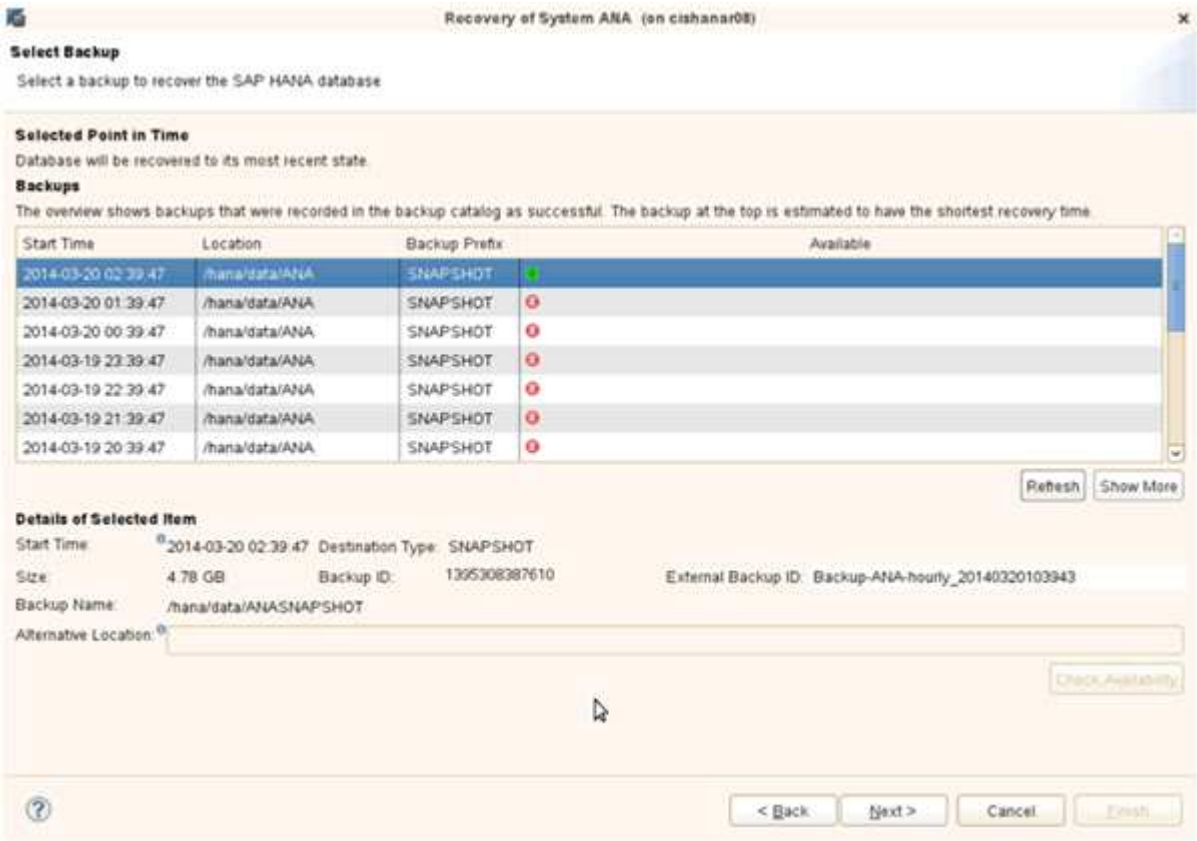
```
mount -o remount /hana/data/ANA/mnt00001
mount -o remount /hana/data/ANA/mnt00002
mount -o remount /hana/data/ANA/mnt00003
```

17. □到 SAP HANA Studio 并□□ * 刷新 * 以更新可用□□列表。

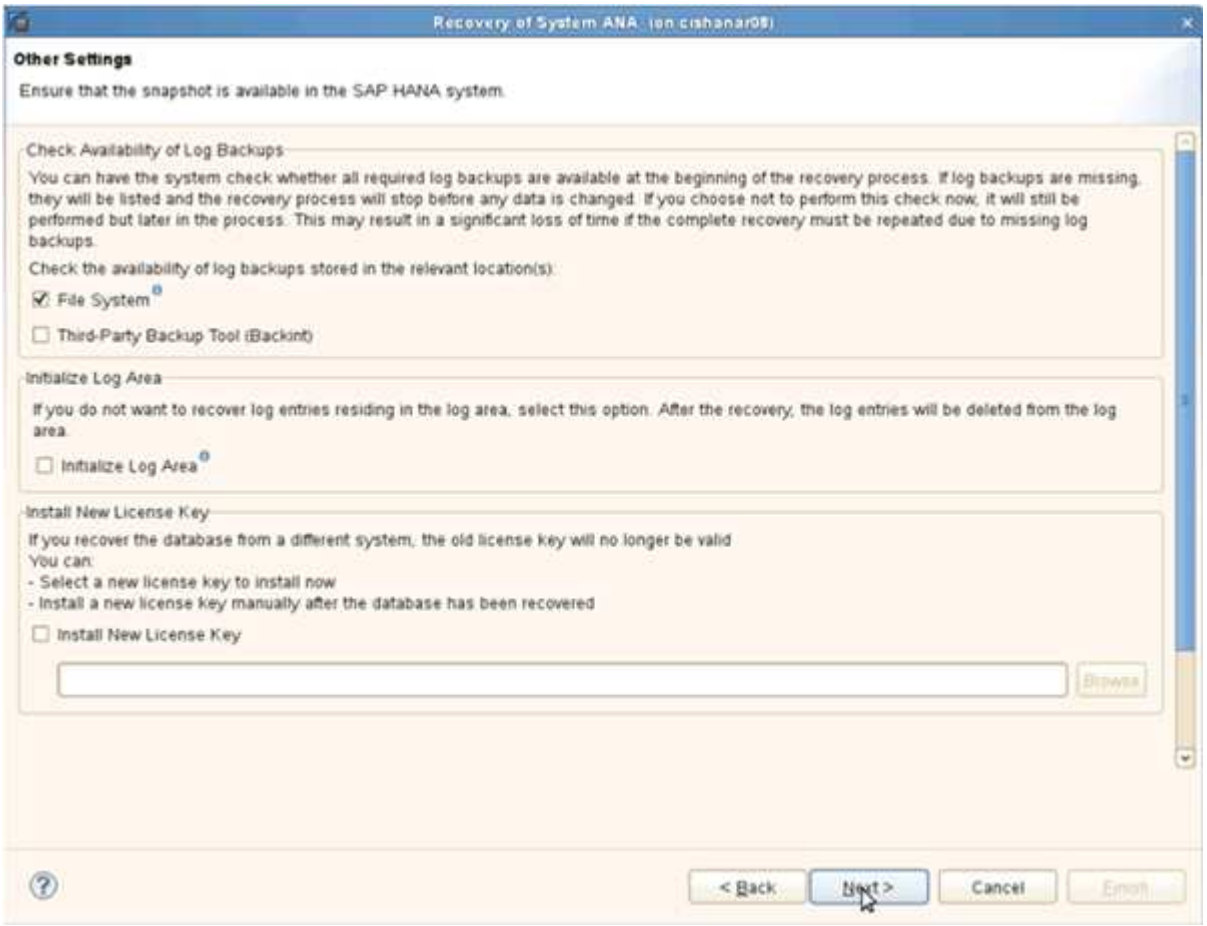


使用 Snap Creator □原的□□在□□列表中□示□色□□。

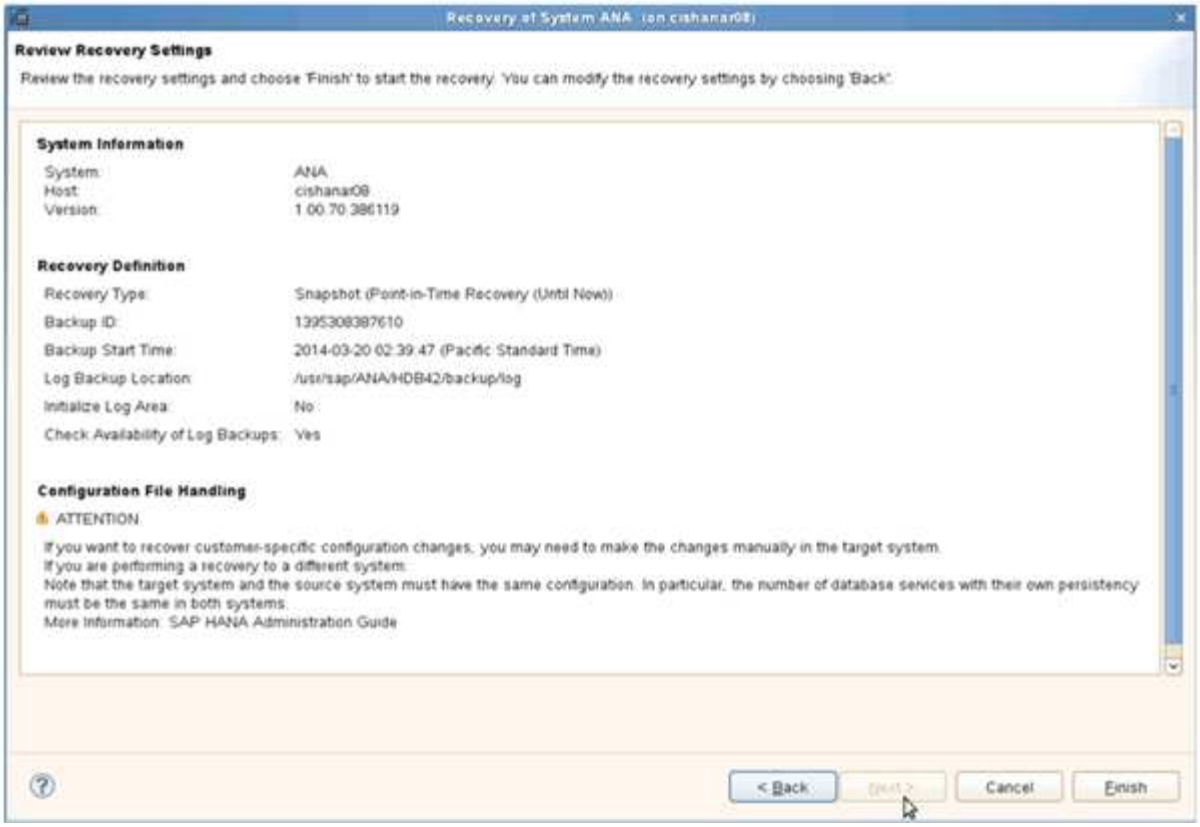
18. □□□□并□□ * 下一□ * 。



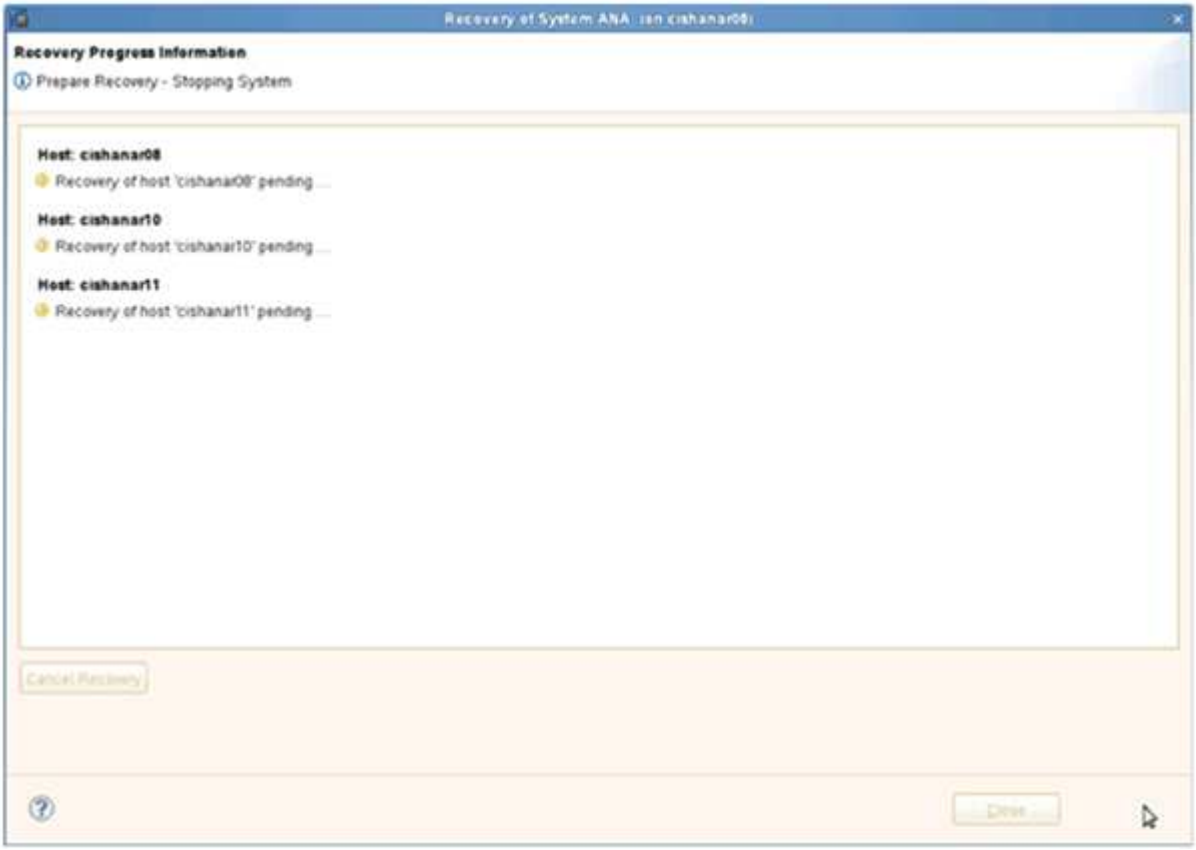
19. 根据需要□□其他□置，然后□□ * 下一□ * 。



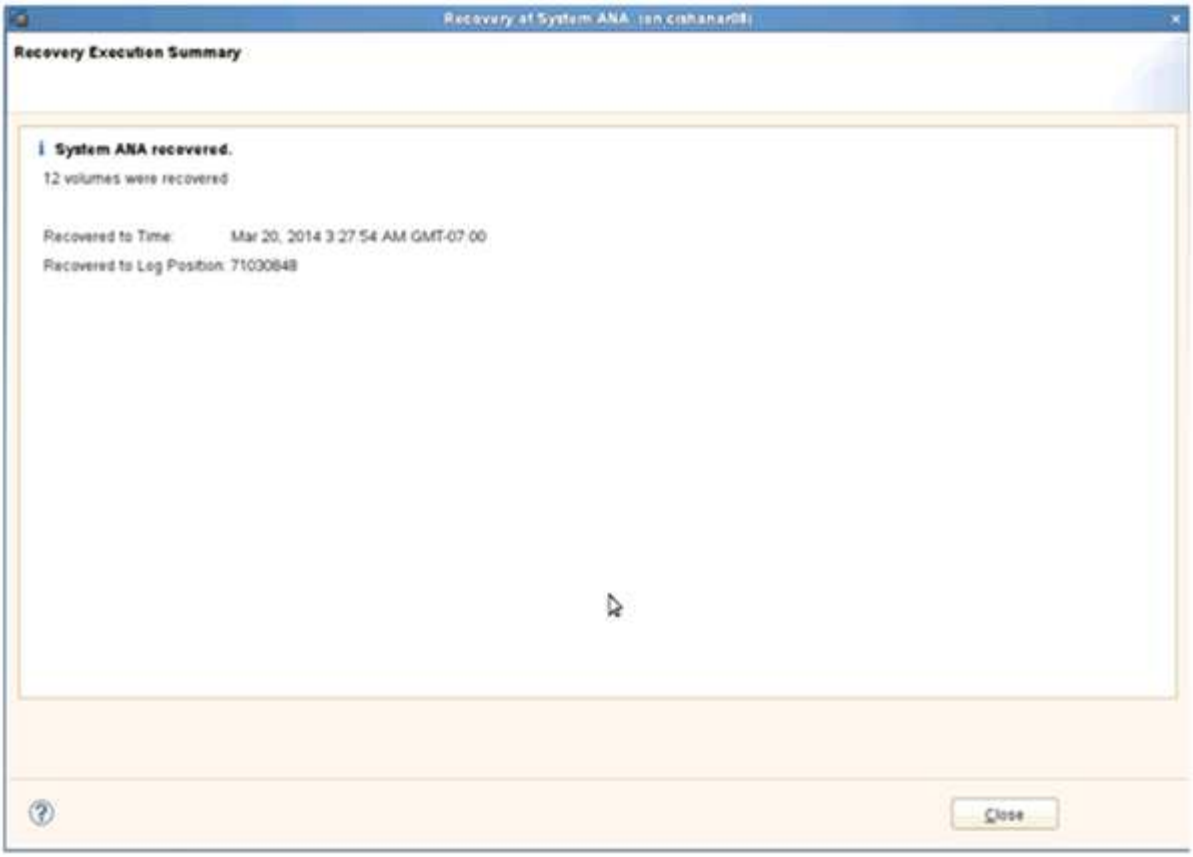
20. □□ * 完成 * 。



恢□□程□始。



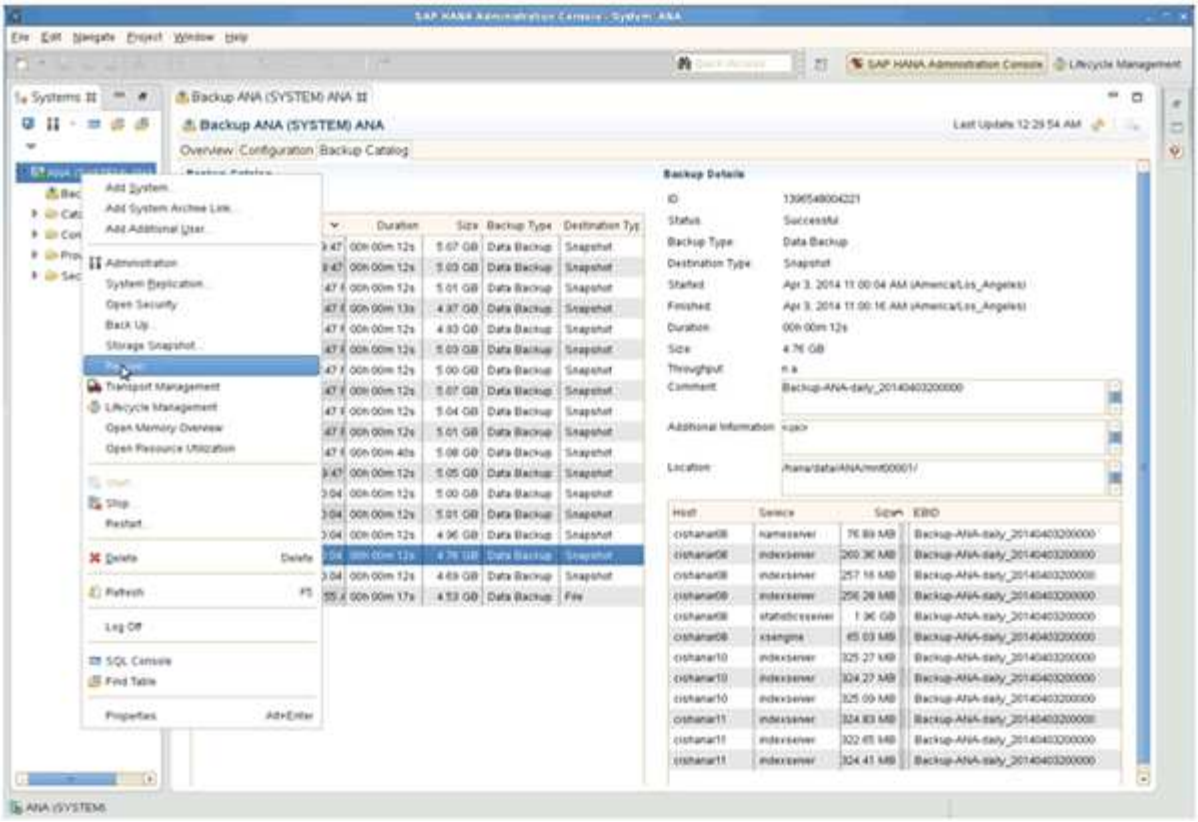
21. 恢□完成后，根据需要恢□ SnapVault □系。



从二□存□□原和恢□数据□

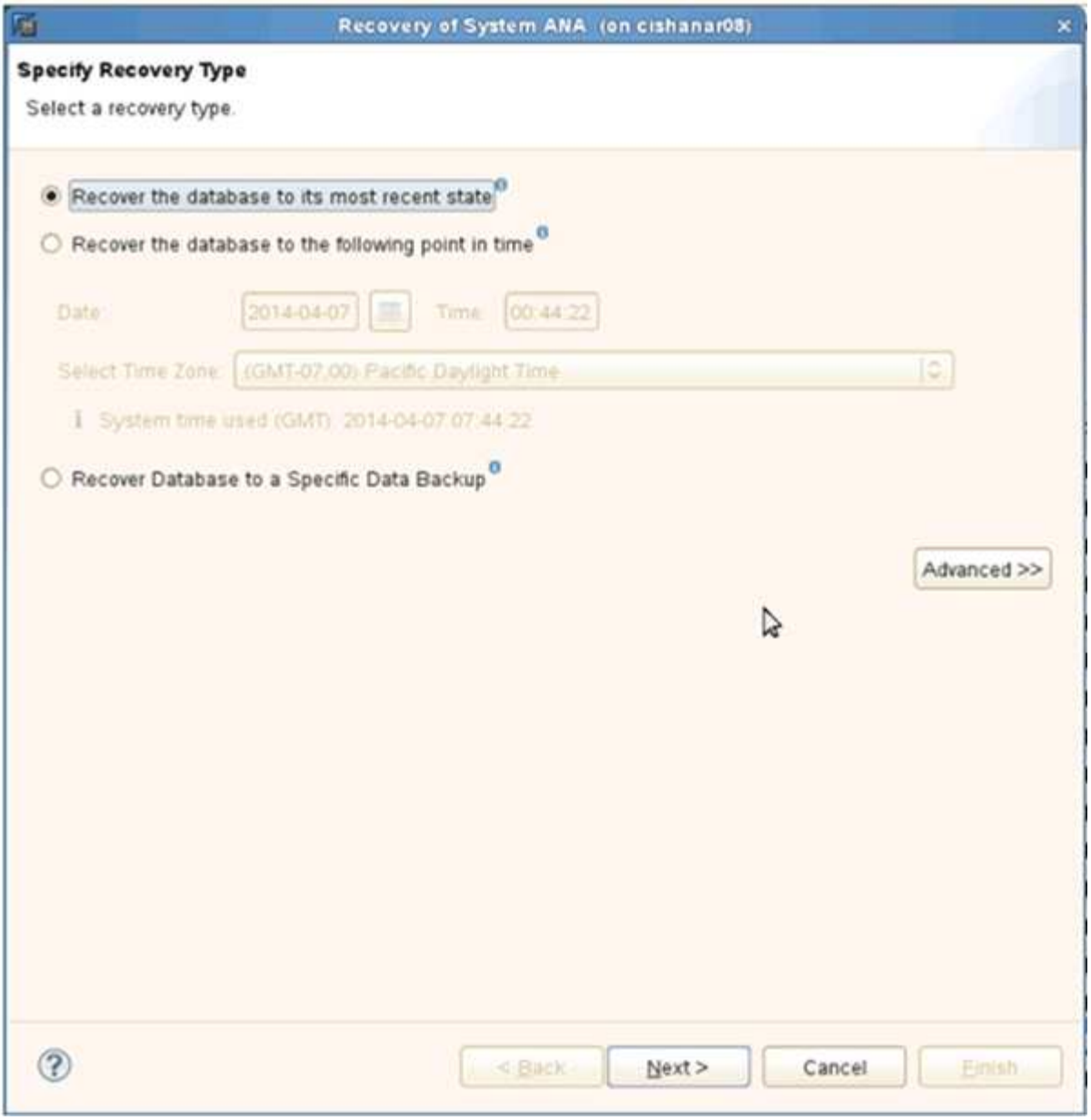
□可以从二□存□□原和恢□数据□。

1. 在 SAP HANA Studio 中，□ SAP HANA 系□□□ * 恢□ * 。

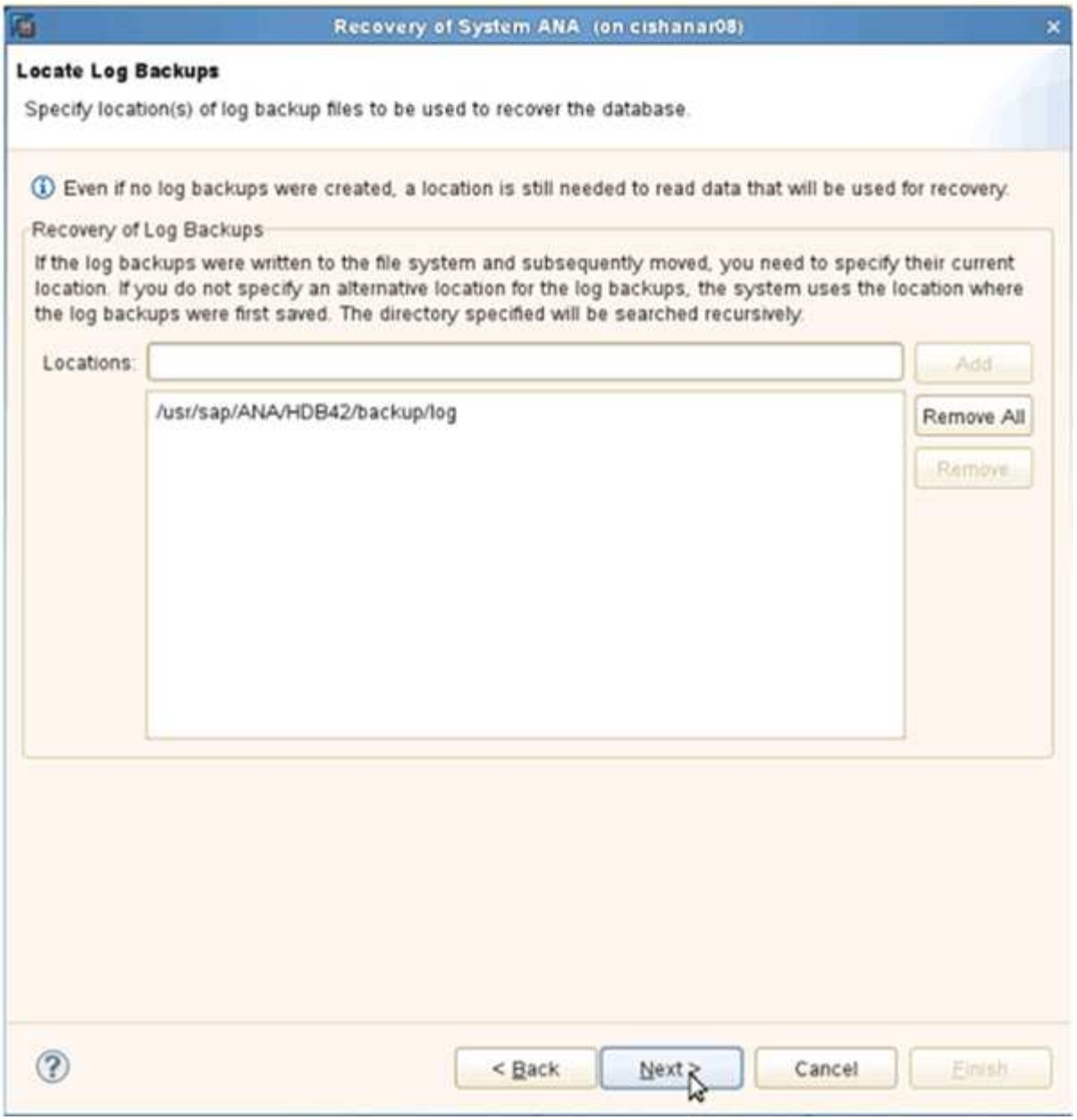


SAP HANA 系□将□□。

2. □□恢□□型并□□ * 下一□ * 。

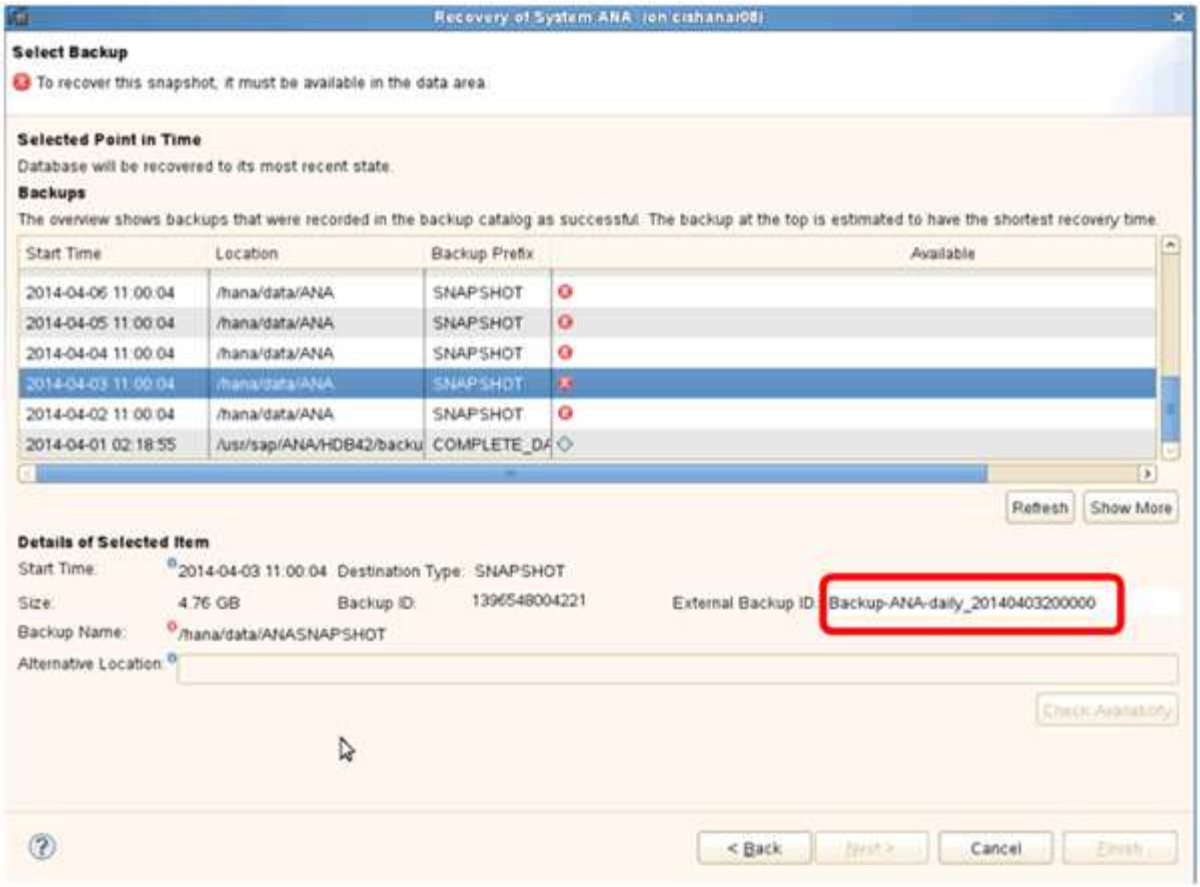


3. 提供日志□□位置，然后□□ * 下一□ * 。

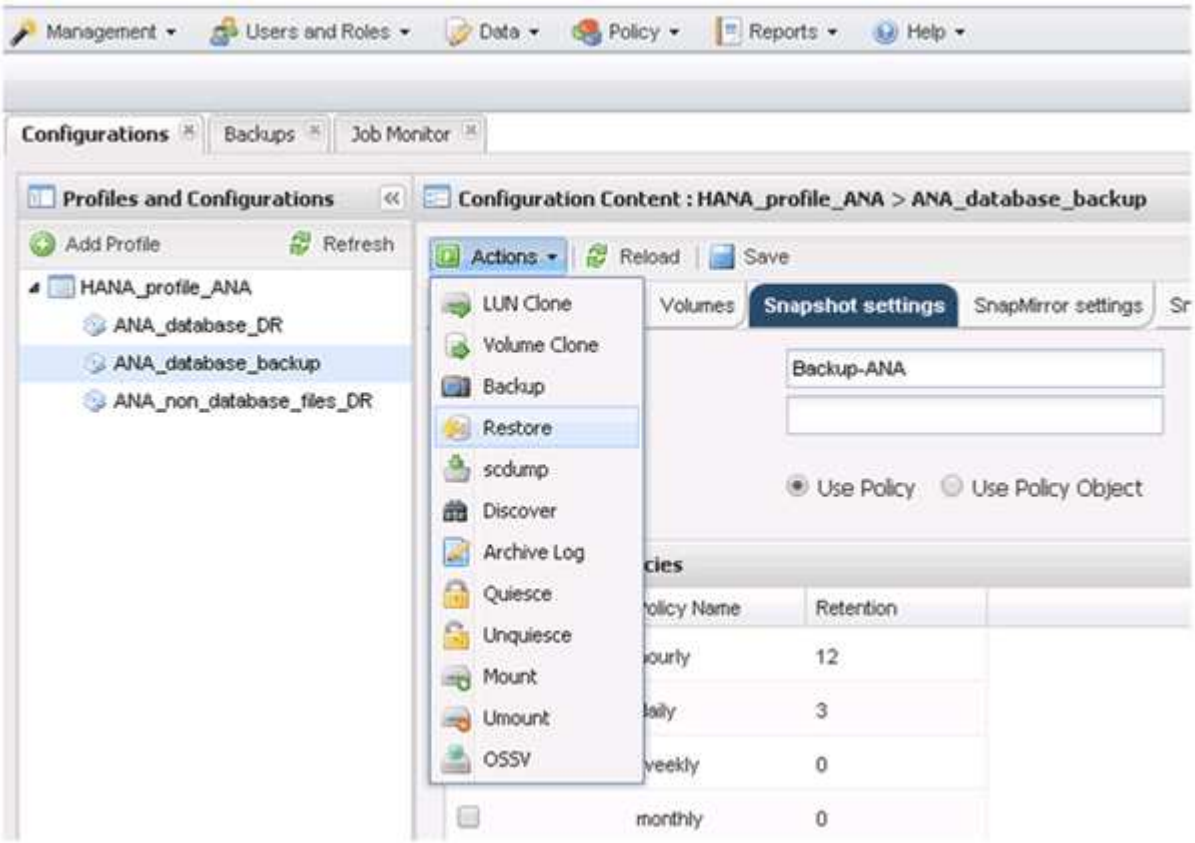


此□将根据□□目□的内容□示可用□□列表。

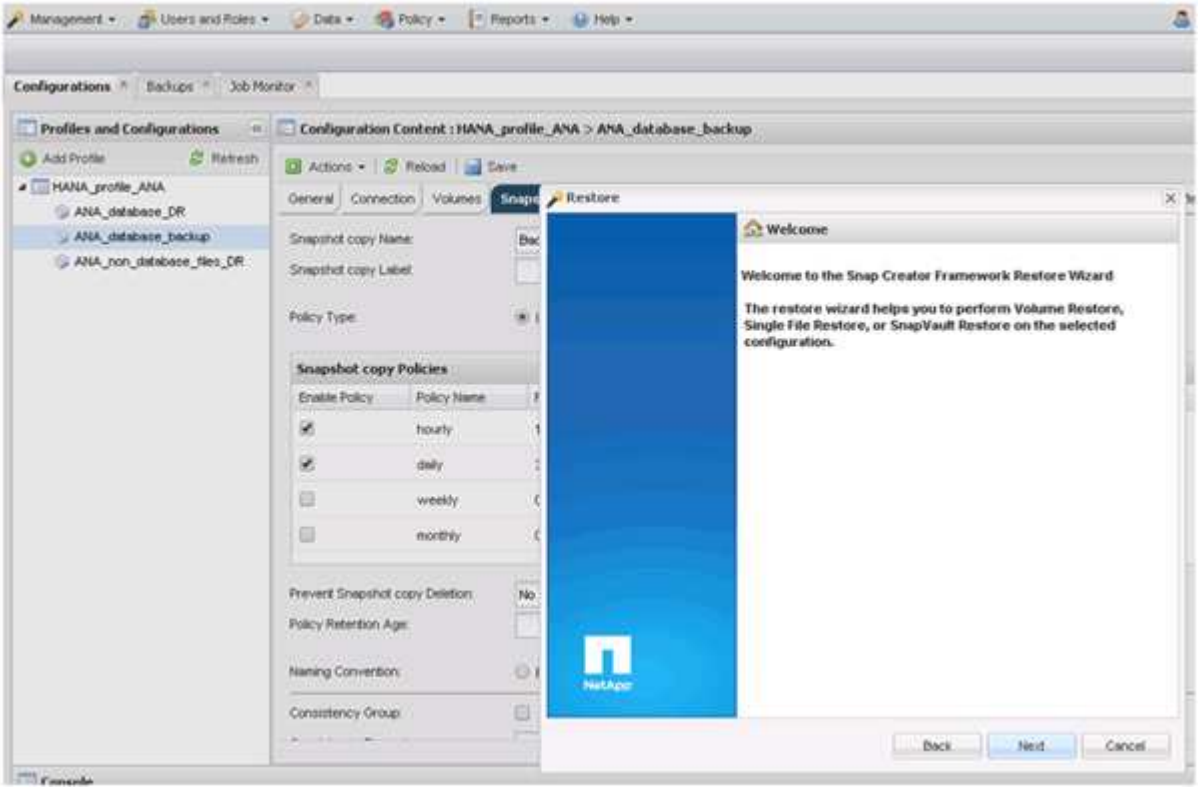
4. □□所需的□□并□下外部□□ ID 。



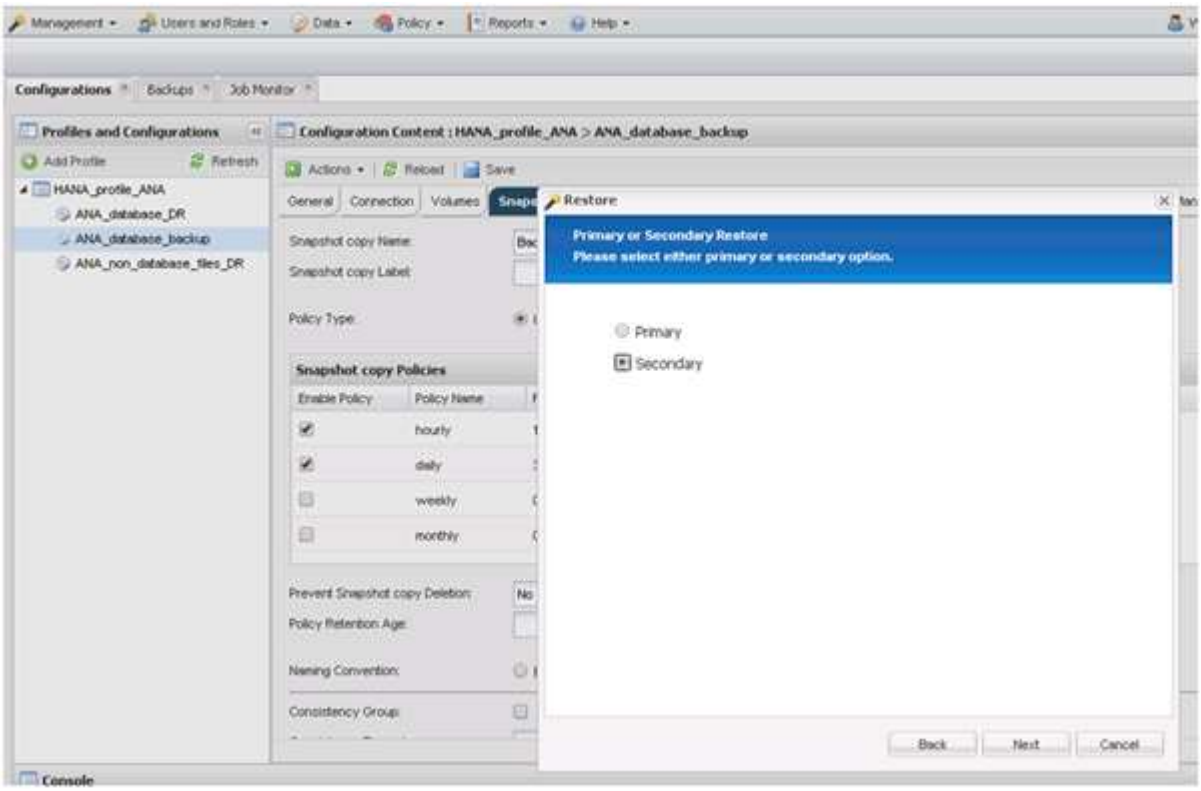
5. □到 Snap Creator GUI 。
6. □□ SAP HANA 系□，然后□□ * 操作 * > * □原 * 。



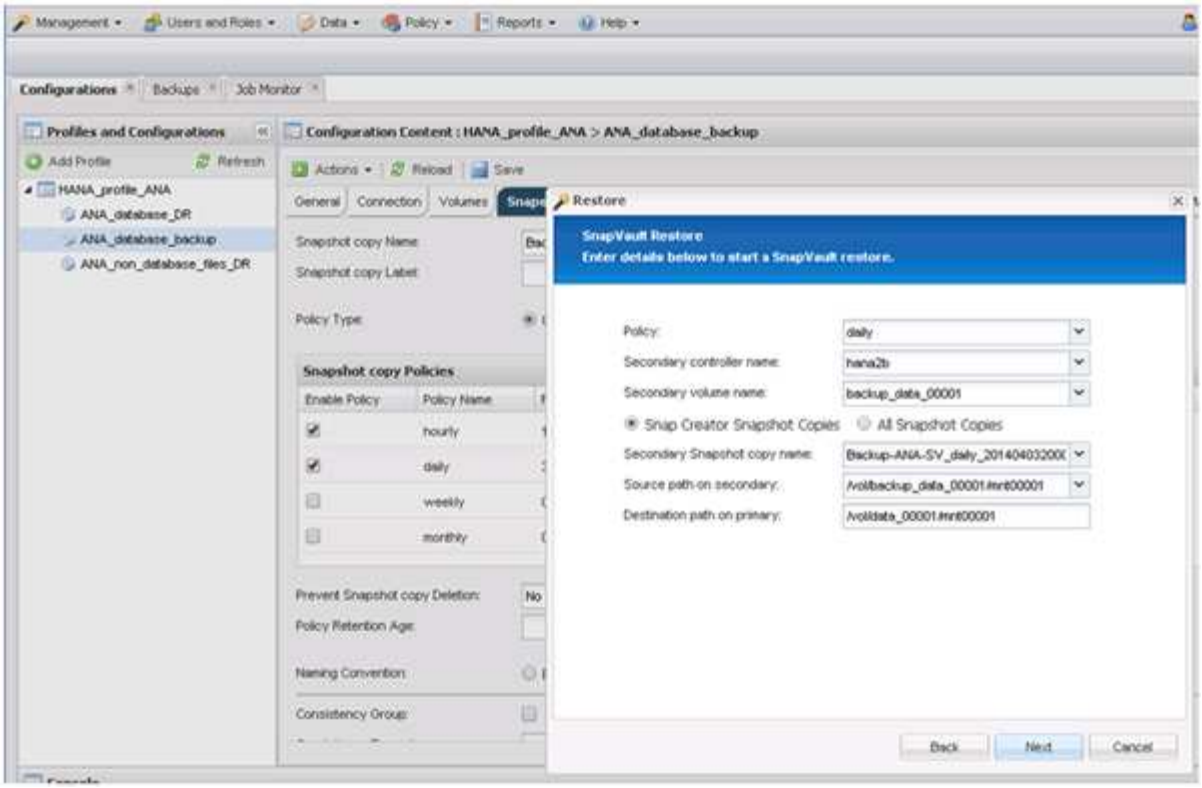
此□将□示□迎屏幕。



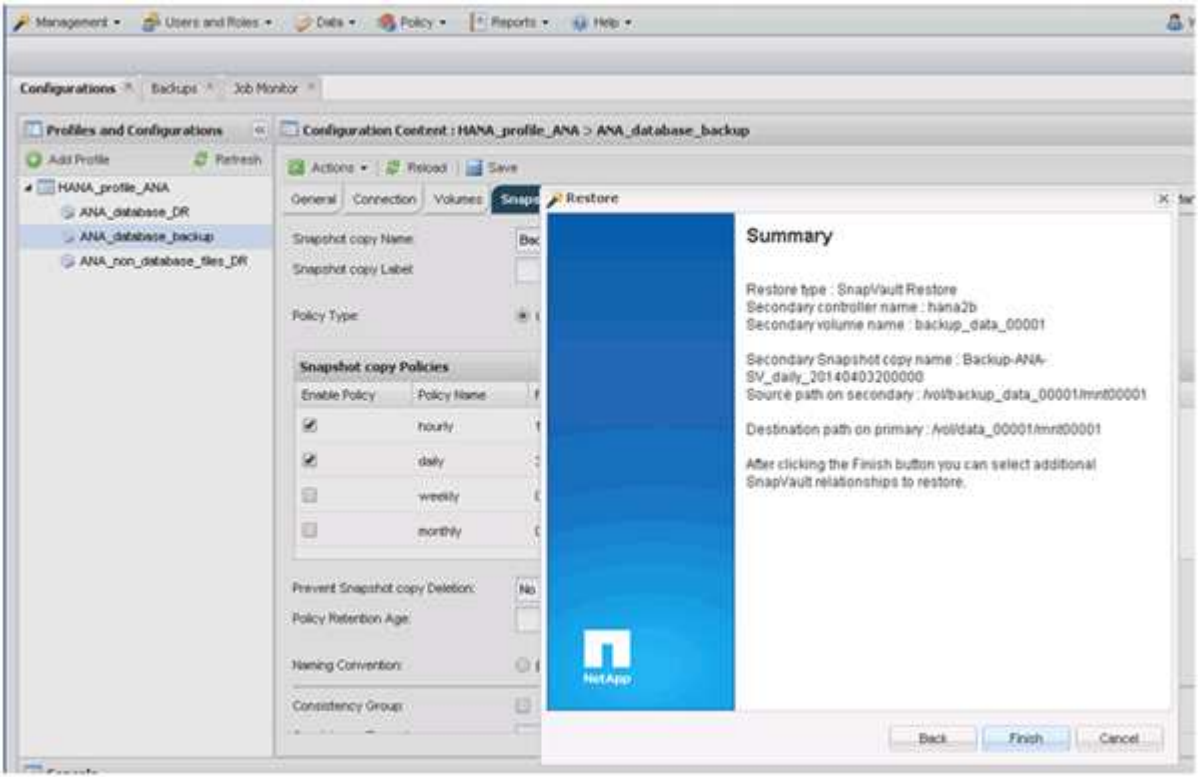
7. □□ * 下一□ * 。
8. □□ * 二□ * 并□□ * 下一□ * 。



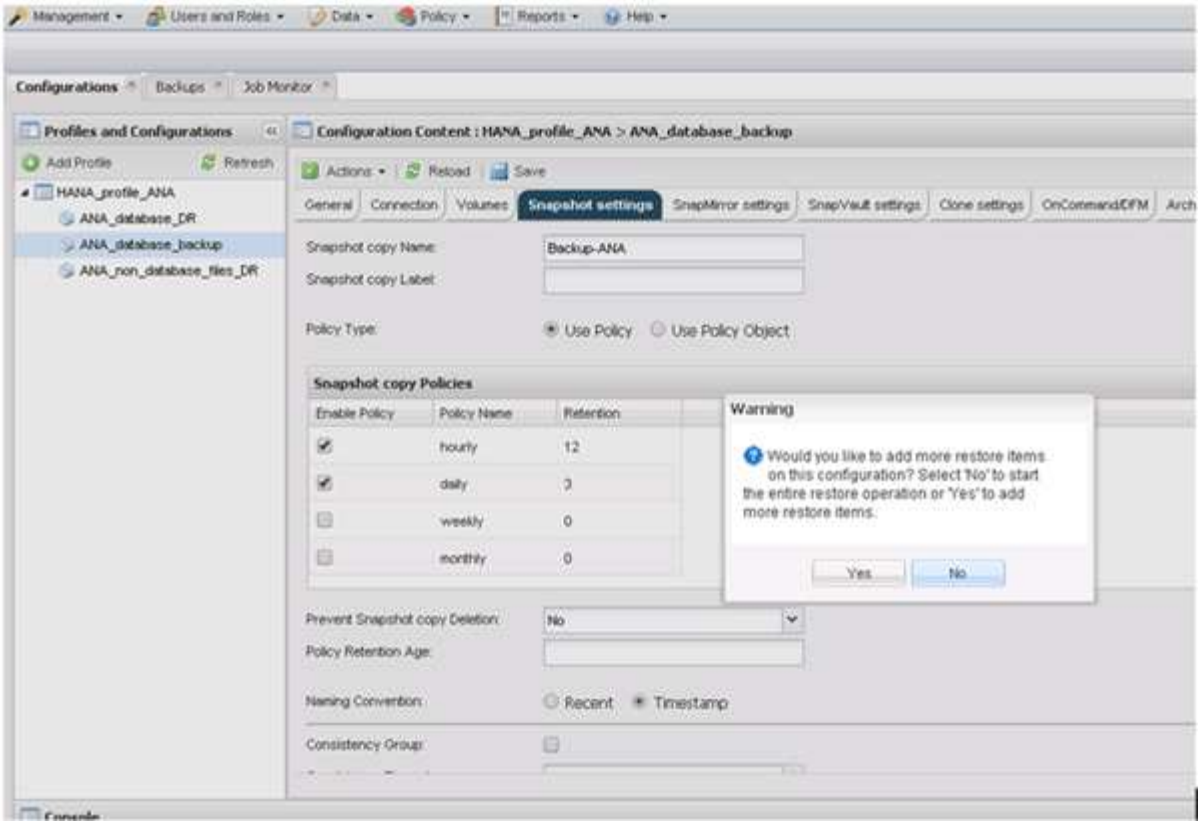
9. □入所需信息。Snapshot 名称与在 SAP HANA Studio 中□□的□□ ID □□。



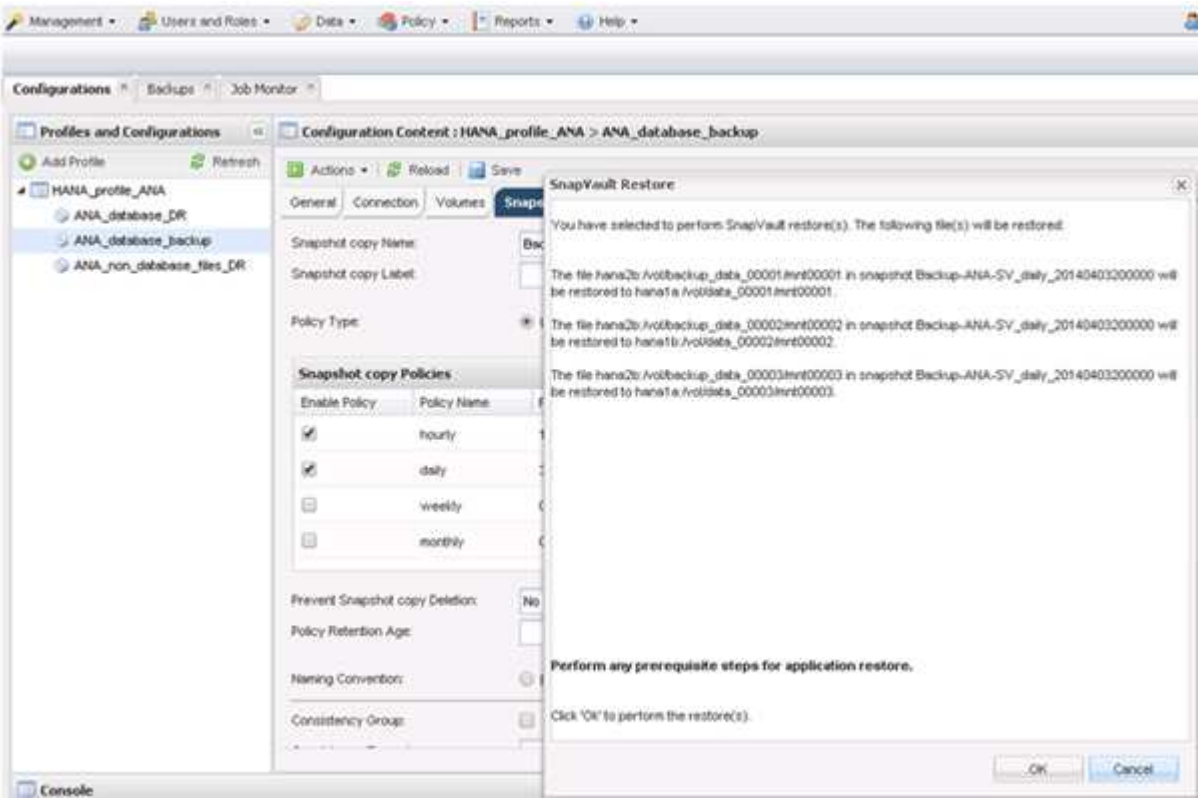
10. □□ * 完成 * 。



11. □□ * 是 * 添加更多□以□行□原。



12. □需要□原的所有卷提供所需信息。在□置 data_00001 中，需要□□原□程□□ data_00002 和 data_00003 。



13. □□所有卷后，□□ * □定 * 以□□□原□程。

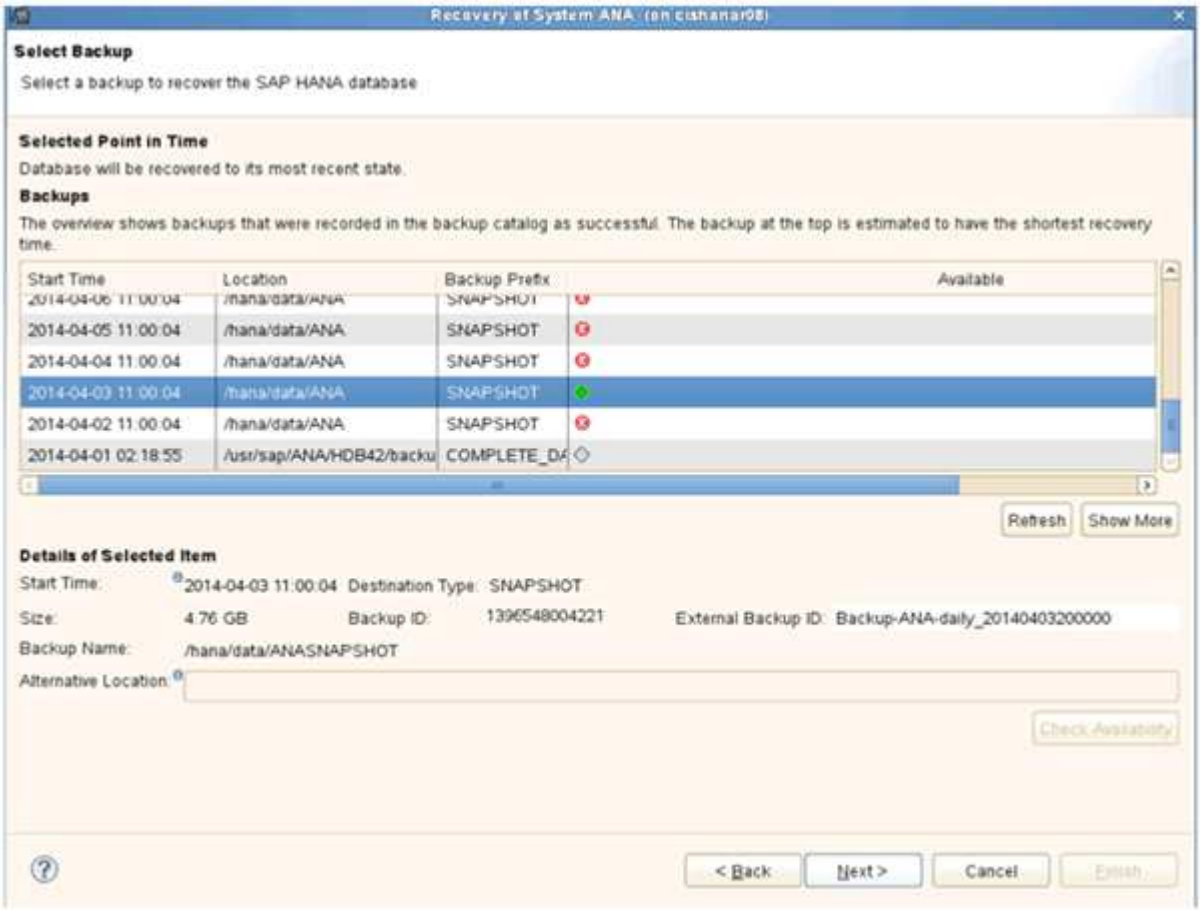
□等待□原□程完成。

14. 在□个数据□□点上，重新挂□所有数据卷以清理 " SNFS 句柄 " 。

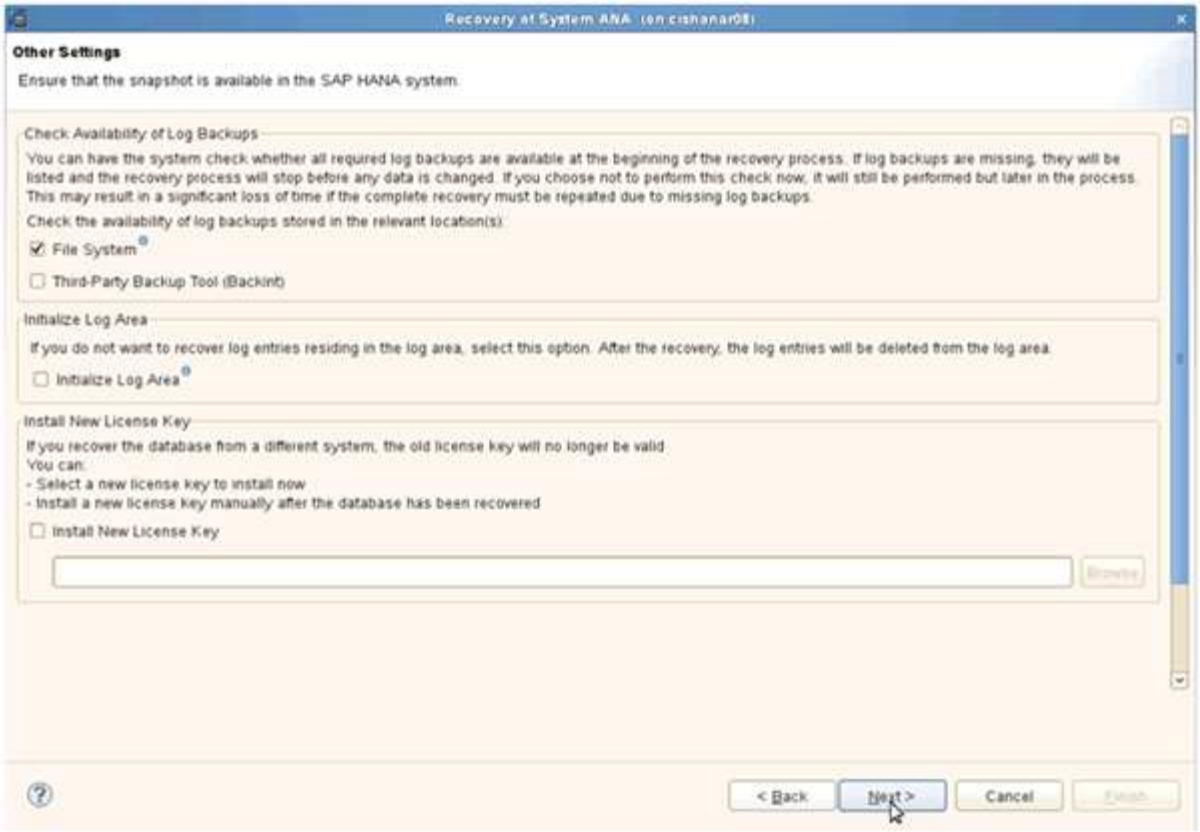
    在此示例中，所有三个卷都需要在□个数据□□点上重新挂□。

```
mount -o remount /hana/data/ANA/mnt00001
mount -o remount /hana/data/ANA/mnt00002
mount -o remount /hana/data/ANA/mnt00003
```

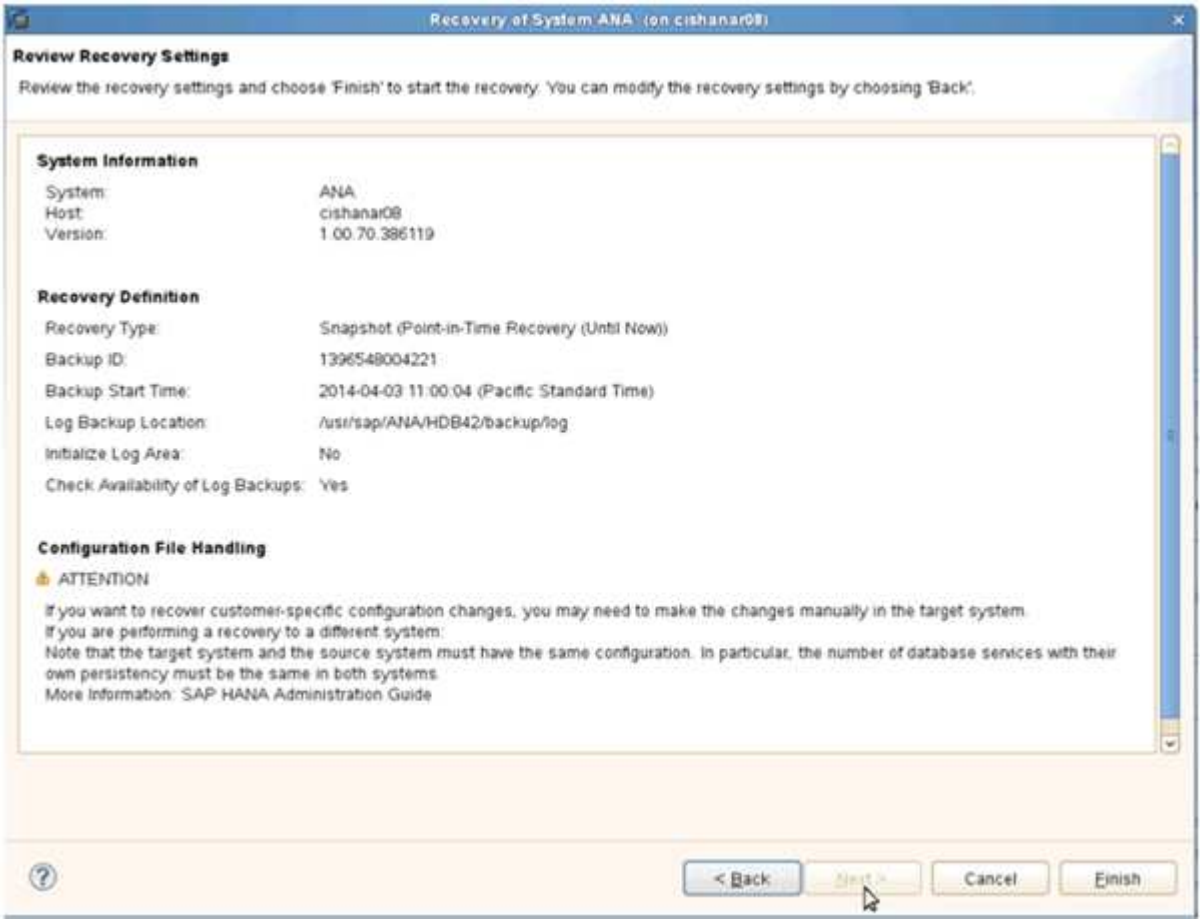
15. □到 SAP HANA Studio 并□□ * 刷新 * 以更新□□列表。



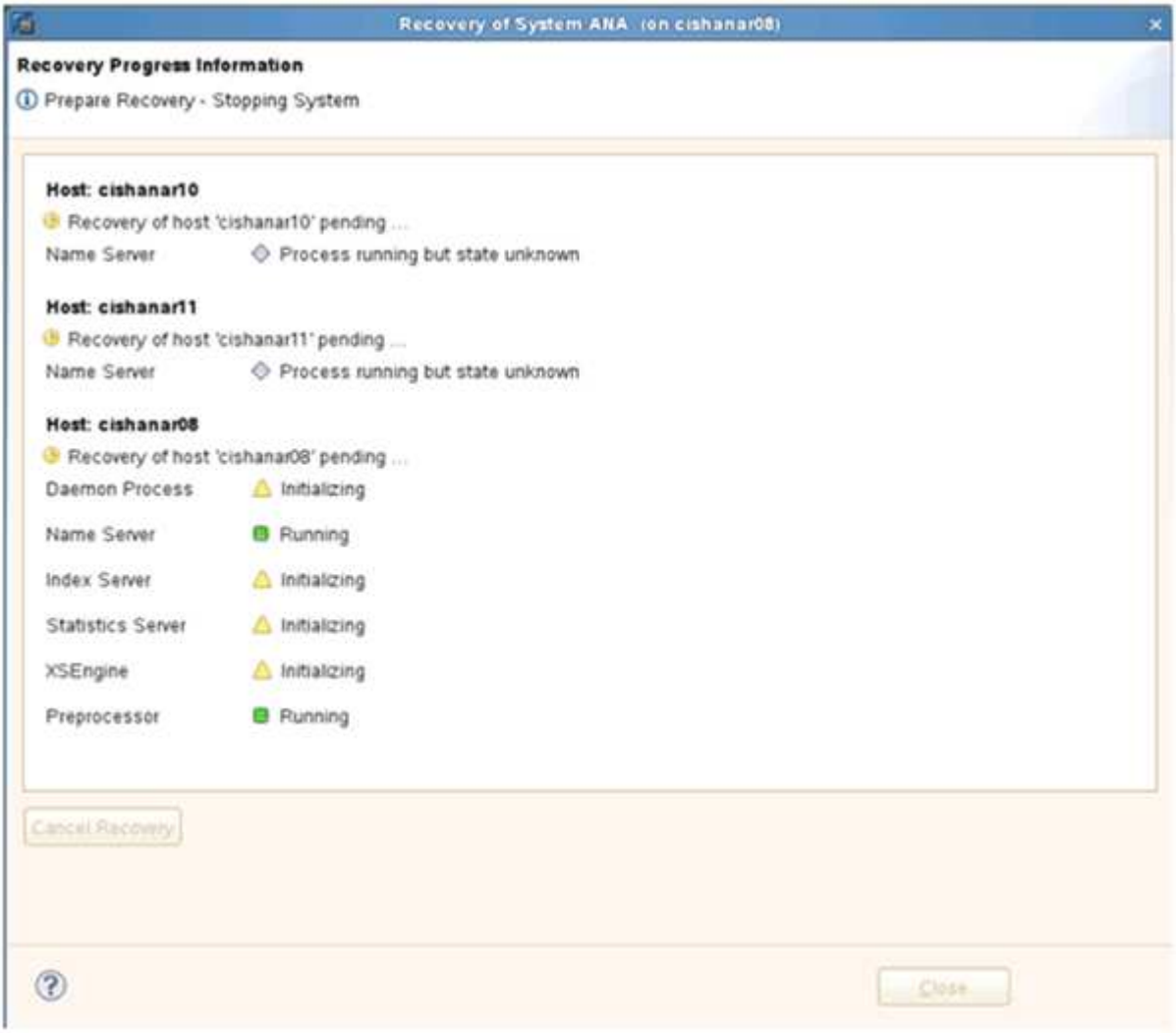
16. 使用 Snap Creator □原的□□在□□列表中□示□色□□。□□□□并□□ * 下一□ * 。
17. 根据需要□□其他□置，然后□□ * 下一□ * 。


Recovery of System ANA (on cishanar08)
Other Settings
Ensure that the snapshot is available in the SAP HANA system.
Check Availability of Log Backups
You can have the system check whether all required log backups are available at the beginning of the recovery process. If log backups are missing, they will be listed and the recovery process will stop before any data is changed. If you choose not to perform this check now, it will still be performed but later in the process. This may result in a significant loss of time if the complete recovery must be repeated due to missing log backups.
Check the availability of log backups stored in the relevant location(s):
File System
Third-Party Backup Tool (Backint)
Initialize Log Area
If you do not want to recover log entries residing in the log area, select this option. After the recovery, the log entries will be deleted from the log area.
Initialize Log Area
Install New License Key
If you recover the database from a different system, the old license key will no longer be valid.
You can:
- Select a new license key to install now
- Install a new license key manually after the database has been recovered
Install New License Key
Browse...
< Back
Next >
Cancel
Finish

18. □□ * 完成 * 。


Recovery of System ANA (on cishanar08)
Review Recovery Settings
Review the recovery settings and choose 'Finish' to start the recovery. You can modify the recovery settings by choosing 'Back'.
System Information
System: ANA
Host: cishanar08
Version: 1.00.70.386119
Recovery Definition
Recovery Type: Snapshot (Point-in-Time Recovery (Until Now))
Backup ID: 1396548004221
Backup Start Time: 2014-04-03 11:00:04 (Pacific Standard Time)
Log Backup Location: /usr/sap/ANA/HDB42/backup/log
Initialize Log Area: No
Check Availability of Log Backups: Yes
Configuration File Handling
ATTENTION
If you want to recover customer-specific configuration changes, you may need to make the changes manually in the target system.
If you are performing a recovery to a different system:
Note that the target system and the source system must have the same configuration. In particular, the number of database services with their own persistency must be the same in both systems.
More Information: SAP HANA Administration Guide
< Back
Next >
Cancel
Finish

恢□□程□始。



19. 恢□□程完成后，根据需要恢□ SnapVault □系。

□原后恢□ SnapVault □系

如果未使用最新的 Snapshot □□□行任何□原，□会□除主存□系□上的 SnapVault □系。

□原和恢□□程完成后，必□恢□ SnapVault □系，以便可以使用 Snap Creator 再次□行□□。否□， Snap Creator 将□□描述□示一条□□消息，因□它在主存□系□上□不到 SnapVault □系。

如果源卷和目□卷之□仍存在通用 Snapshot 副本，□所需的数据□□将基于□量□□。

恢□与 7- 模式 Data ONTAP 的 SnapVault □系

如果使用最新的 Snapshot □□以外的 Snapshot □□□行□原，□需要恢□ SnapVault □系，以便 Snap Creator 可以□□□行□□。

1. □入以下命令，恢□与 7- 模式 Data ONTAP 的 SnapVault □系。SnapVault start -r -S source_controller ： source_volumebackup_controller ： backup_volume

   □属于 SAP HANA 数据□的所有卷□行此□□。

```
hana2b> snapvault start -r -S hana1a:/vol/data_00001/mnt00001
hana2b:/vol/backup_data_00001/mnt00001
The resync base snapshot will be: Backup-ANA-SV_daily_20140406200000
Resync may alter the data in this qtree.
Are you sure you want to resync the qtree? y
Mon Apr  7 14:08:21 CEST [hana2b:replication.dst.resync.success:notice]:
SnapVault resync of
/vol/backup_data_00001/mnt00001 to hana1a:/vol/data_00001/mnt00001 was
successful.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana2b> snapvault start -r -S hana1b:/vol/data_00002/mnt00002
hana2b:/vol/backup_data_00002/mnt00002
The resync base snapshot will be: Backup-ANA-SV_daily_20140406200000
Resync may alter the data in this qtree.
Are you sure you want to resync the qtree? y
Mon Apr  7 14:09:49 CEST [hana2b:replication.dst.resync.success:notice]:
SnapVault resync of
/vol/backup_data_00002/mnt00002 to hana1b:/vol/data_00002/mnt00002 was
successful.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana2b> snapvault start -r -S hana1a:/vol/data_00003/mnt00003
hana2b:/vol/backup_data_00003/mnt00003
The resync base snapshot will be: Backup-ANA-SV_daily_20140406200000
Resync may alter the data in this qtree.
Are you sure you want to resync the qtree? y
Mon Apr  7 14:10:25 CEST [hana2b:replication.dst.resync.success:notice]:
SnapVault resync of
/vol/backup_data_00003/mnt00003 to hana1a:/vol/data_00003/mnt00003 was
successful.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

数据□□完成后，□可以再次使用 Snap Creator □□□□。

恢□与集群模式 **Data ONTAP** 的 **SnapVault** □系

如果使用最新的 Snapshot □□以外的 Snapshot □□□行□原，□需要恢□ SnapVault □系，以便 Snap Creator 可以□□□行□□。

1. 重新□建并重新同□ SnapVault □系。

```
hana::> snapmirror create -source-path hana1a:hana_data -destination
-path
hana2b:backup_hana_data -type XDP
Operation succeeded: snapmirror create the relationship with destination
hana2b:backup_hana_data.

hana::> snapmirror resync -destination-path hana2b:backup_hana_data
-type XDP

Warning: All data newer than Snapshot copy sc-backup-
daily_20140430121000 on volume
hana2b:backup_hana_data will be deleted.
Do you want to continue? {y|n}: y
[Job 6554] Job is queued: initiate snapmirror resync to destination
"hana2b:backup_hana_data".
[Job 6554] Job succeeded: SnapMirror Resync Transfer Queued
```

2. 要□□重新□□ SnapVault □□，需要手□□建 Snapshot 副本。

```
hana::> snapshot create -vserver hana1a -volume hana_data -snapshot
sv_resync

hana::> snapshot modify -vserver hana1a -volume hana_data -snapshot
sv_resync -snapmirror-label daily

hana::> snapmirror update -destination-path hana2b:backup_hana_data
Operation is queued: snapmirror update of destination
hana2b:backup_hana_data.
```

3. □□ SnapVault □系是否□示在目□列表中。

```
hana::> snapmirror list-destinations -source-path hana1a:hana_data
                                                      Progress
Source             Destination         Transfer   Last
Relationship
Path         Type  Path         Status Progress   Updated      Id
----------- ----- ------------ ------- --------- ------------
---------------
hana1a:hana_data
            XDP   hana2b:backup_hana_data
                                Transferring
                                        38.46KB   04/30 18:15:54
                                                                  9137fb83-
cba9-11e3-85d7-123478563412
```

在主存□□生故障后□原数据□

在主存□□生故障或从主存□的卷中□除所有 Snapshot 副本后， Snap Creator 将无法□理□原，因□主存□系□上将不再存在 SnapVault □系。

使用在 7- 模式下□行的 Data ONTAP 在主存□□生故障后□原数据□

在□行 7- 模式 Data ONTAP 的主存□系□出□故障后，□可以□原 SAP HANA 数据□。

1. 在□□情况下，必□使用以下命令在二□存□系□上直接□行□原： SnapVault restore -s snapshot_name -S backup_controller ： backup_volumesource_controller ： source_volume

   □属于 SAP HANA 数据□的所有卷□行此□□。

```
hana1a>  snapvault restore -s Backup-ANA-SV_hourly_20140410103943 -S
hana2b:/vol/backup_data_00001/mnt00001 hana1a:/vol/data_00001/mnt00001
Restore will overwrite existing data in /vol/data_00001/mnt00001.
Are you sure you want to continue? y
Thu Apr 10 11:55:55 CEST [hana1a:vdisk.qtreePreserveComplete:info]:
Qtree preserve is complete for /vol/data_00001/mnt00001.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana1a>  snapvault restore -s Backup-ANA-SV_hourly_20140410103943 -S
hana2b:/vol/backup_data_00003/mnt00003 hana1a:/vol/data_00003/mnt00003
Restore will overwrite existing data in /vol/data_00003/mnt00003.
Are you sure you want to continue? y
Thu Apr 10 11:58:18 CEST [hana1a:vdisk.qtreePreserveComplete:info]:
Qtree preserve is complete for /vol/data_00003/mnt00003.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

```
hana1b>  snapvault restore -s Backup-ANA-SV_hourly_20140410103943 -S
hana2b:/vol/backup_data_00002/mnt00002 hana1b:/vol/data_00002/mnt00002
Restore will overwrite existing data in /vol/data_00002/mnt00002.
Are you sure you want to continue? y
Thu Apr 10 12:01:29 CEST [hana1b:vdisk.qtreePreserveComplete:info]:
Qtree preserve is complete for /vol/data_00002/mnt00002.
Transfer started.
Monitor progress with 'snapvault status' or the snapmirror log.
```

□原□程完成后，□可以使用 SAP HANA □行恢□。

使用集群模式 **Data ONTAP** 在主存□□生故障后□原数据□

□可以在□行集群模式 Data ONTAP 的主存□系□出□故障后□原 SAP HANA 数据□。

假□主卷完全□失，□需要□建一个新的主卷，然后从□□卷□行□原。

1. □建□型□数据保□的主卷。

```
hana::> volume create -vserver hana1a -volume hana_data -aggregate
aggr_sas_101 -size 300G -state online -type DP -policy default -autosize
-mode grow_shrink -space-guarantee none
-snapshot-policy none -foreground true
[Job 6744] Job is queued: Create hana_data.
[Job 6744] Job succeeded: Successful
```

2. 从□□卷□原所有数据。

```
hana::> snapmirror restore -destination-path hana1a:hana_data -source
-path hana2b:backup_hana_data -source-snapshot sc-backup-
daily_20140505121000
[Job 6746] Job is queued: snapmirror restore from source
"hana2b:backup_hana_data" for the
snapshot sc-backup-daily_20140505121000.

hana::> job show -id 6746
Owning
Job ID Name                 Vserver    Node           State
------ -------------------- ---------- -------------- ----------
6746   SnapMirror restore   hana        hana01        Running
       Description: snapmirror restore from source
"hana2b:backup_hana_data" for the snapshot sc-backup-
daily_20140505121000
```

□原□程完成后，□可以使用 SAP HANA □行恢□。

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